Course: IT3021 – Data Warehousing and Business Intelligence

Assignment 02

Student Number – IT23621060

Batch – Kandy UNI

Topic: Cinema Hall Ticket Sales and Customer Behavior

Introduction:

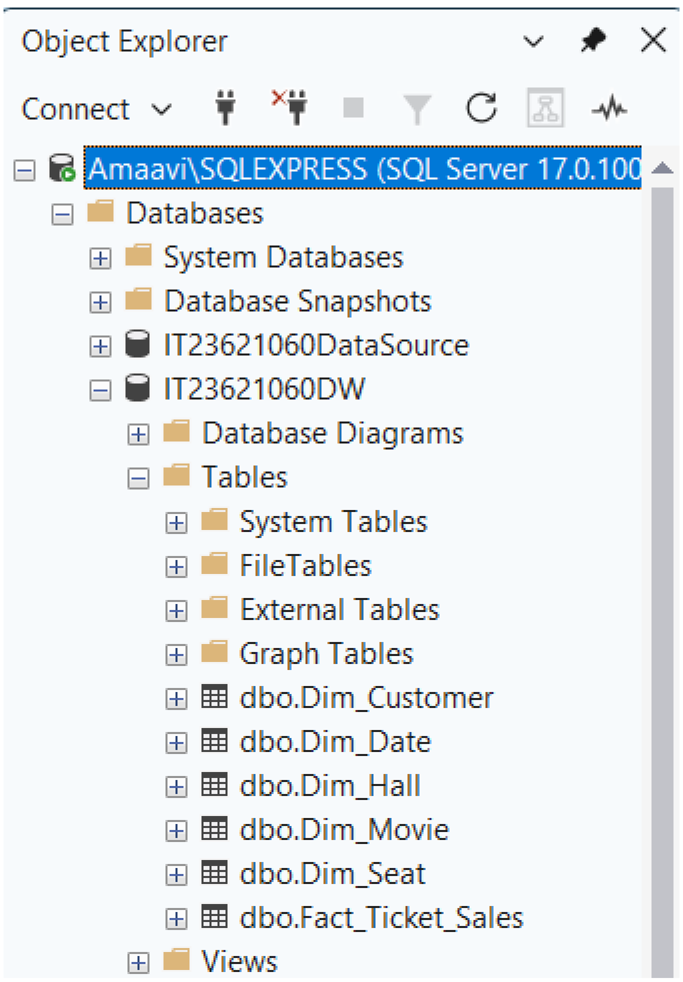
This assignment continues the data warehouse developed in Assignment 1 for the topic Cinema Hall Ticket Sales and Customer Behavior. The existing warehouse was used to implement an SSAS cube, perform OLAP analysis in Excel, and develop interactive reports in Power BI. The final solution supports analytical reporting on cinema ticket sales, halls, movies, dates, and customer-related behavior.

Step 1: Data source for the assignment 2

The data source used for Assignment 2 is the SQL Server data warehouse created in Assignment 1. It follows a star schema design with Fact_Ticket_Sales as the central fact table and Dim_Customer, Dim_Movie, Dim_Hall, Dim_Seat, and Dim_Date as dimensions. The warehouse stores transaction measures such as amount, quantity, and transaction process time, which are used for OLAP analysis and reporting.

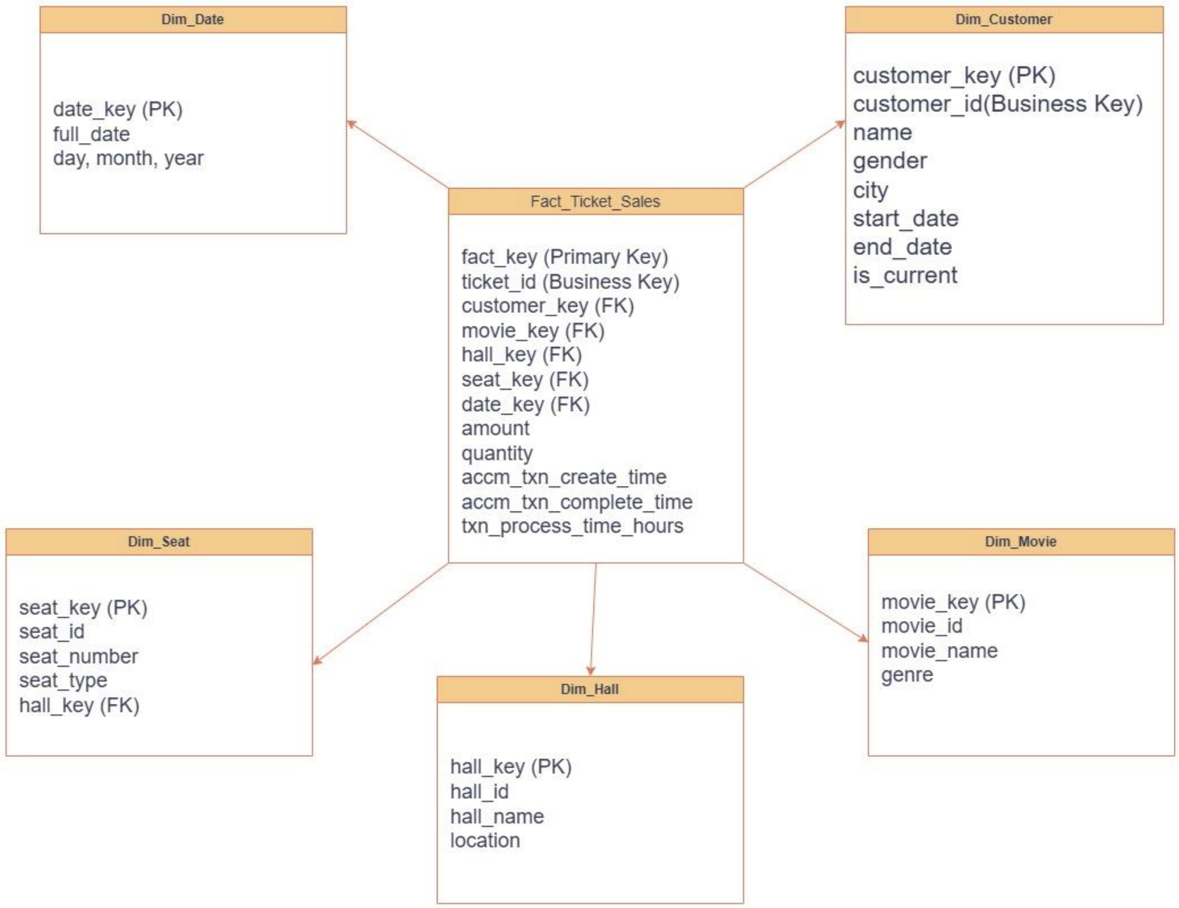
<u>DW database</u>

- Assignment 1 data warehouse used as the source for Assignment 2.



Star schema

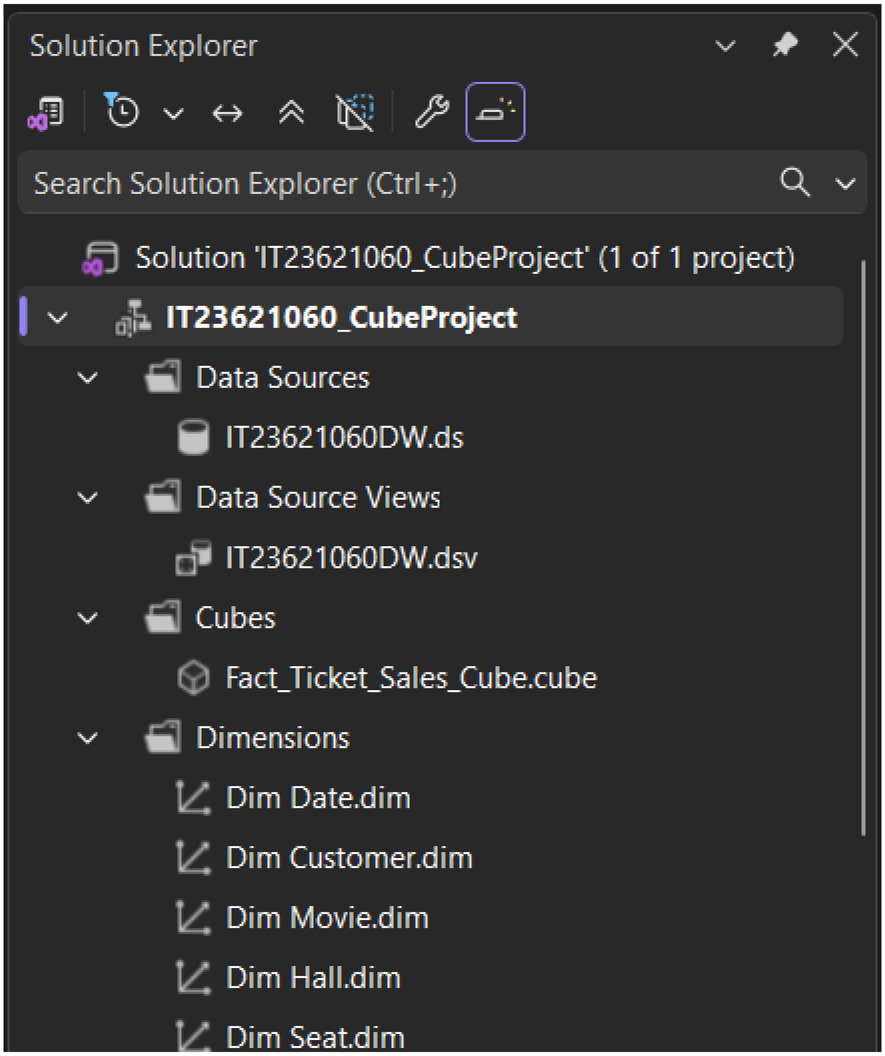
- Star schema of the cinema ticket sales data warehouse.



Step 2: SSAS Cube implementation

2.1. SSAS Cube implementation

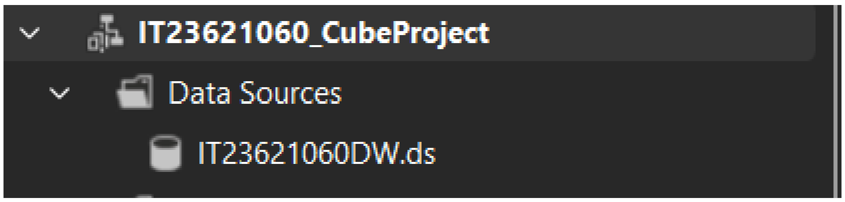
A new Analysis Services Multidimensional Project was created in Visual Studio. The SQL Server Analysis Services instance installed on the local machine was used as the deployment target.

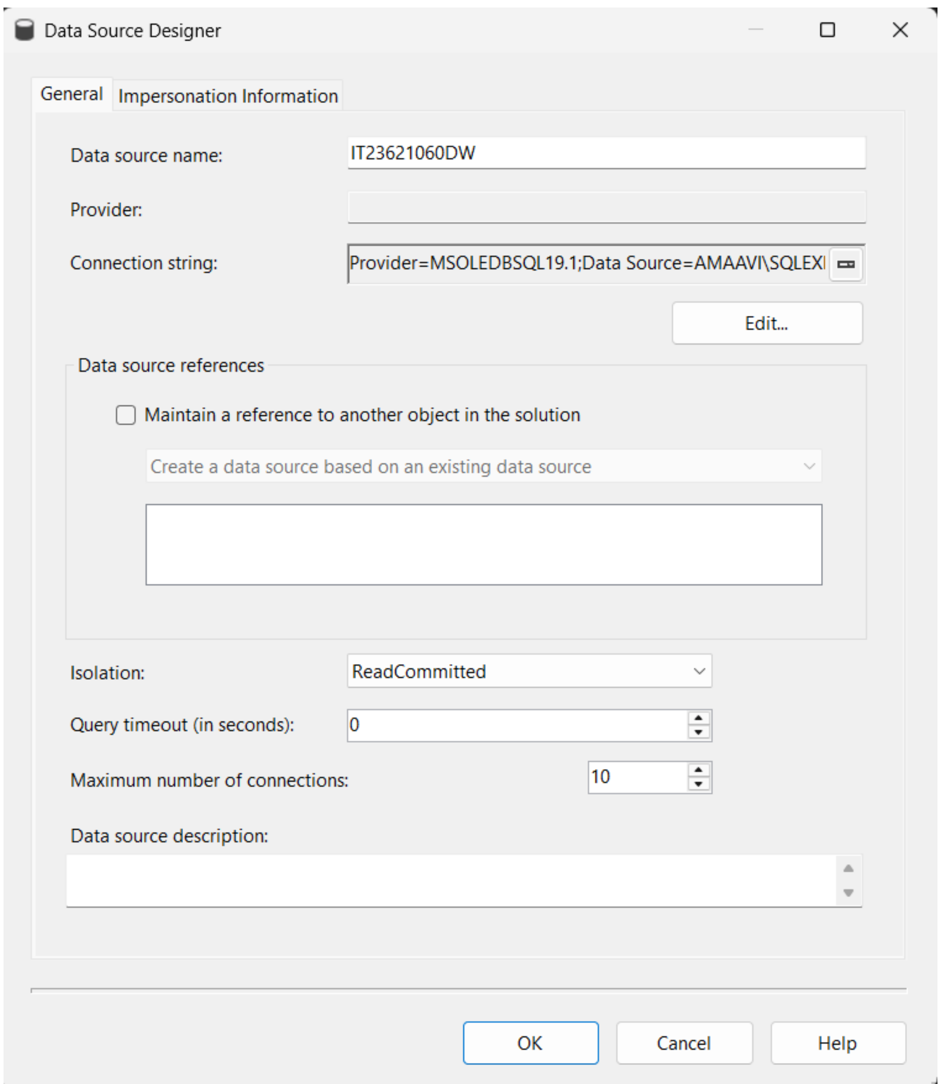


Analysis Services Multidimensional Project created in Visual Studio.

2.2. Creating the Data Source

A data source was created by connecting the SSAS project to the SQL Server database engine containing the IT23621060DW warehouse. Windows Authentication was used and the existing warehouse database was selected.

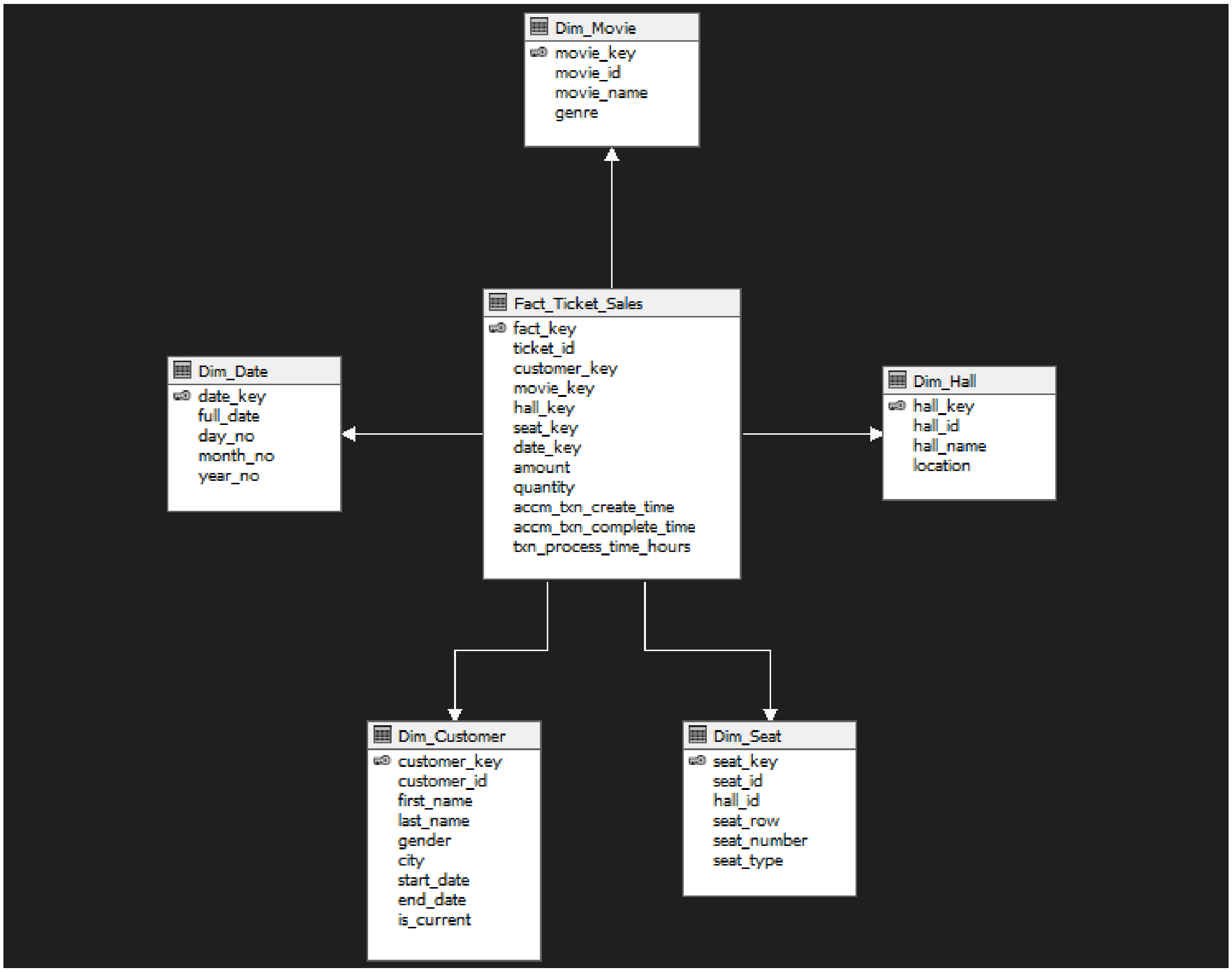


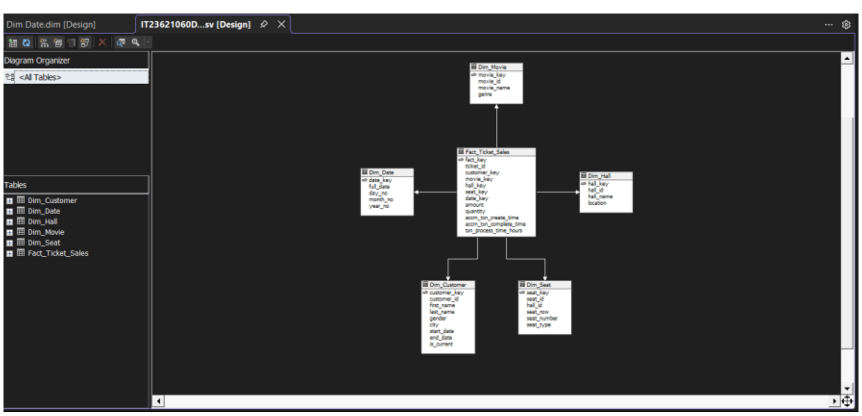


Data source connection to the IT23621060DW warehouse database

2.3. Creating the Data Source View

A Data Source View was created using the fact and dimension tables from the warehouse. The fact table Fact_Ticket_Sales was linked with the dimensions through surrogate keys.



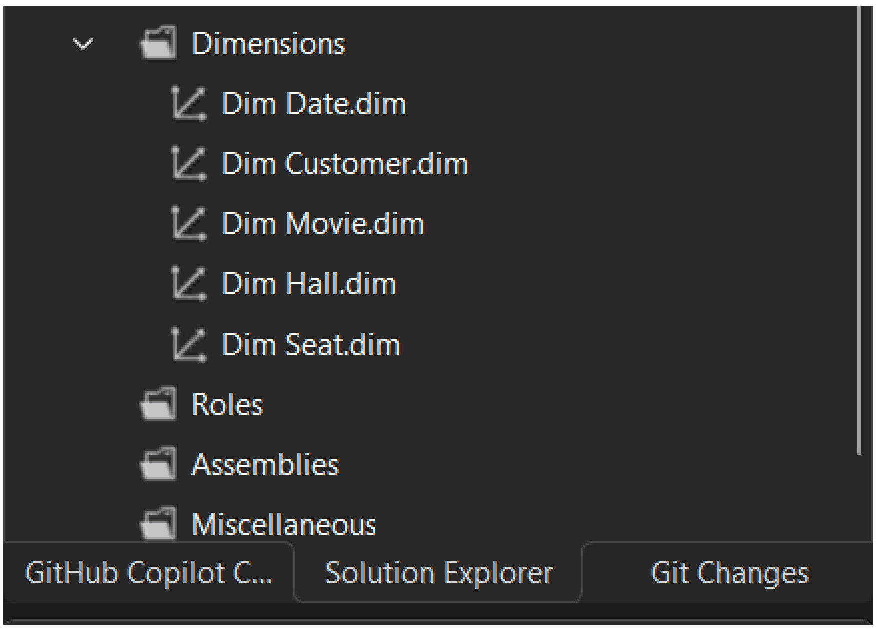


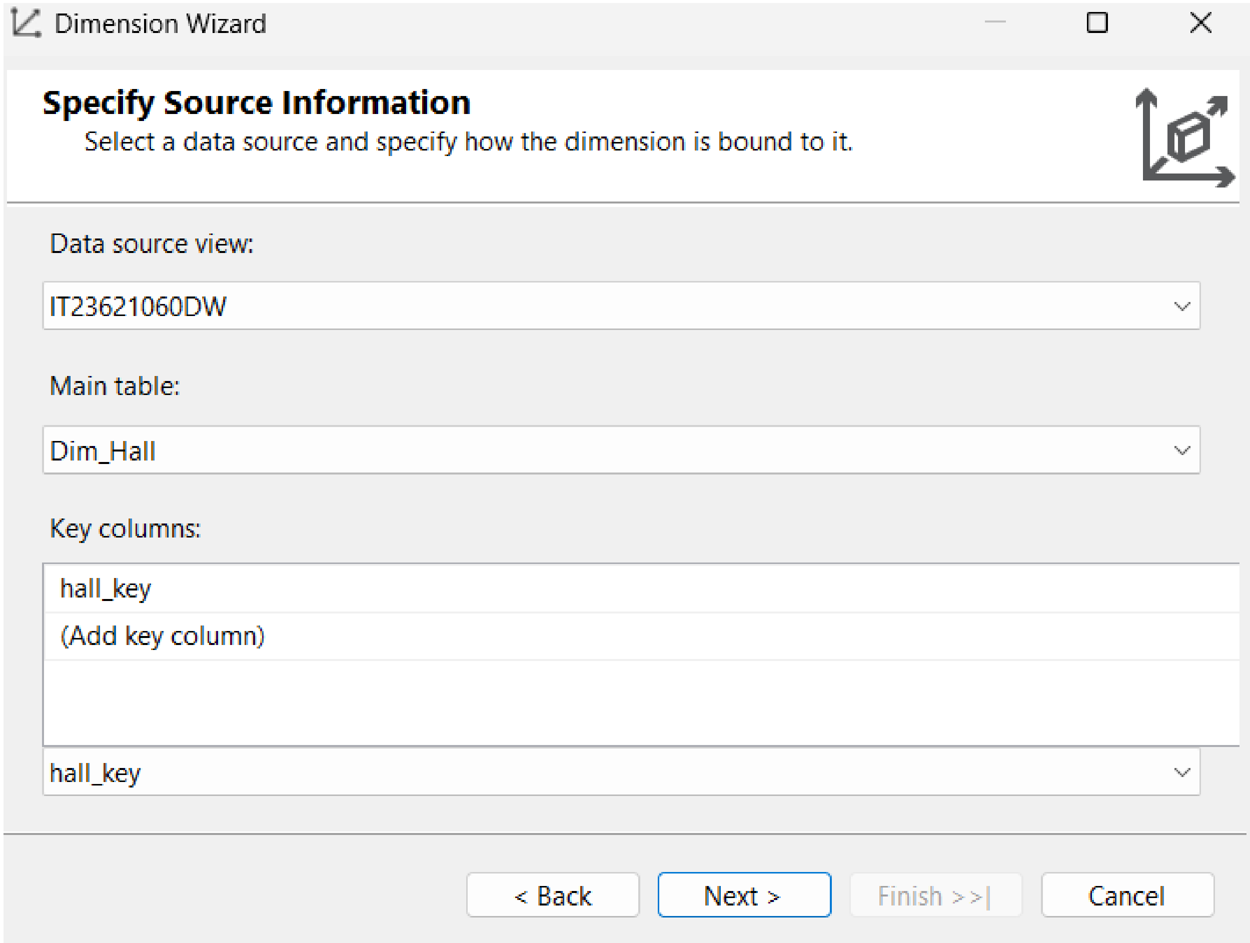
Data Source View showing the star schema relationships.

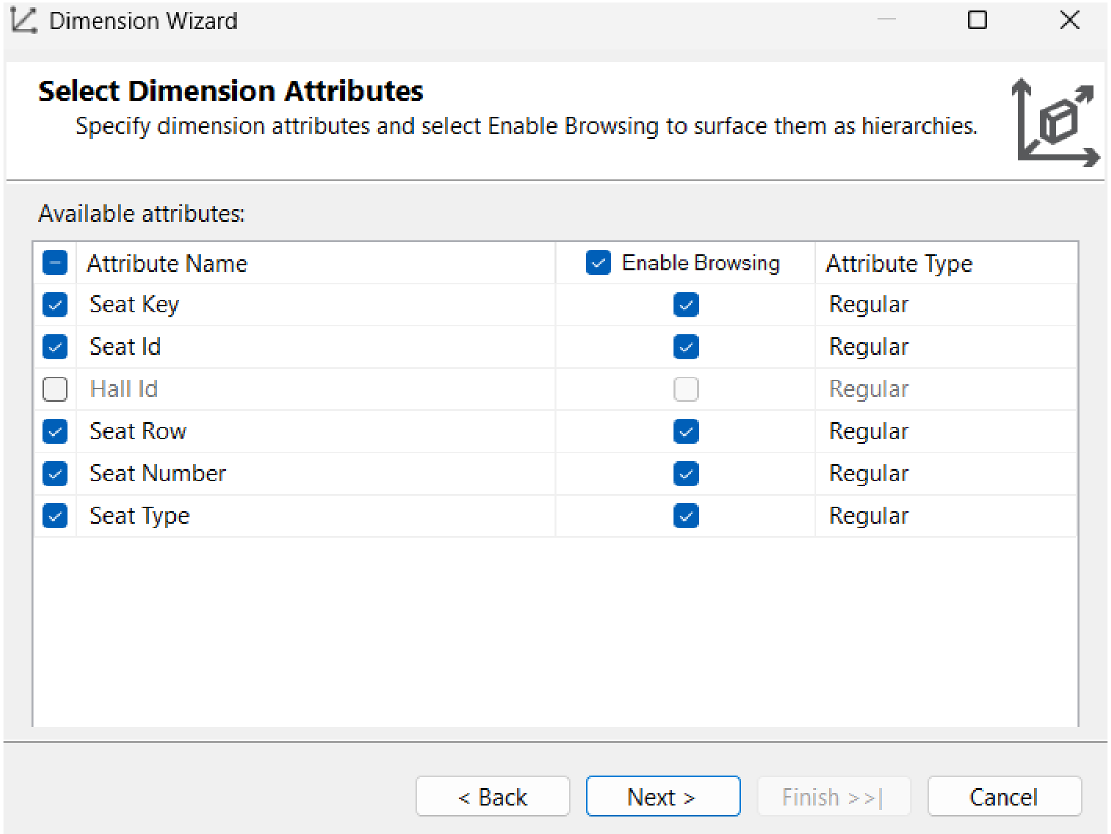
2.4. Creating dimensions

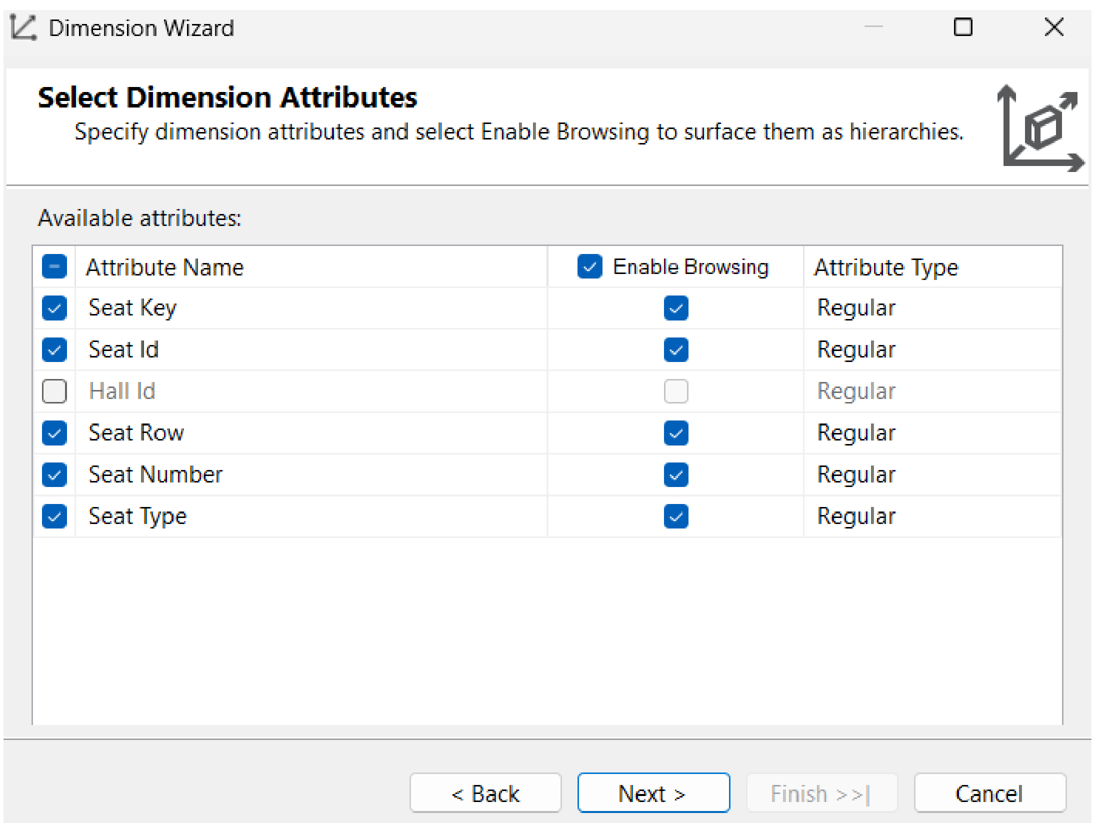
Dimensions were created from the existing dimension tables. User-friendly business attributes were selected for browsing, while technical ETL columns were excluded where not needed.

- Dim_Customer → customer details
- Dim_Movie → movie details
- Dim_Hall → hall details
- Dim_Seat → seat details
- Dim_Date → date analysis

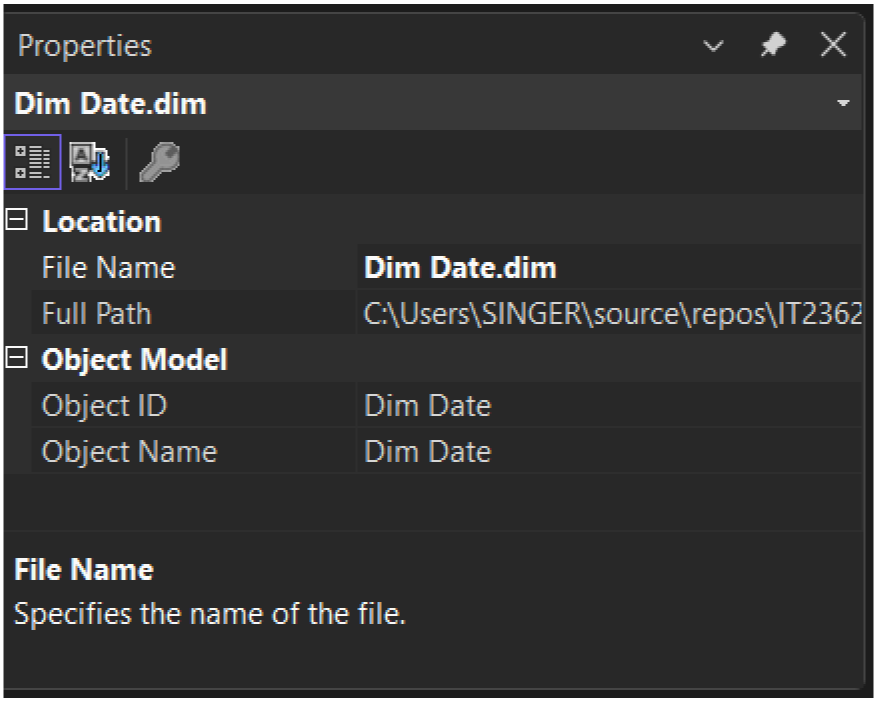


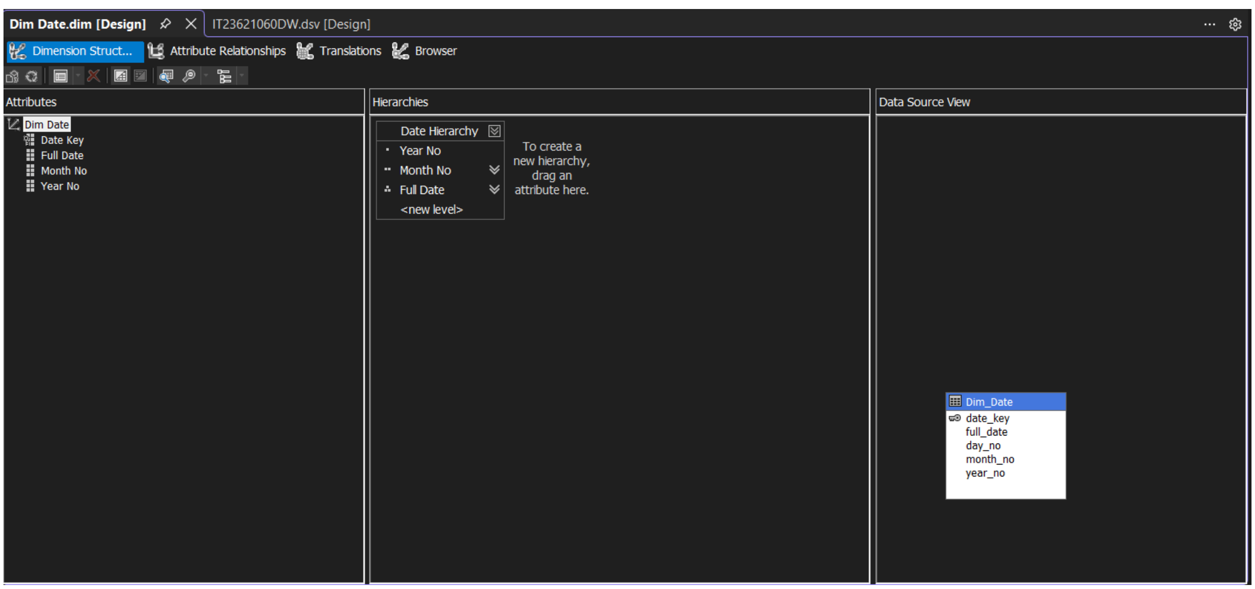

Dimension Wizard
Specify Source Information
Select a data source and specify how the dimension is bound to it.
Data source view:
IT23621060DW
Main table:
Dim_Hall
Key columns:
hall_key
(Add key column)
hall_key
< Back
Next >
Finish >>|
Cancel


Dimension Wizard
Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.
Available attributes:
Attribute Name
Enable Browsing
Attribute Type
Seat Key
Regular
Seat Id
Regular
Hall Id
Regular
Seat Row
Regular
Seat Number
Regular
Seat Type
Regular
< Back
Next >
Finish >>|
Cancel

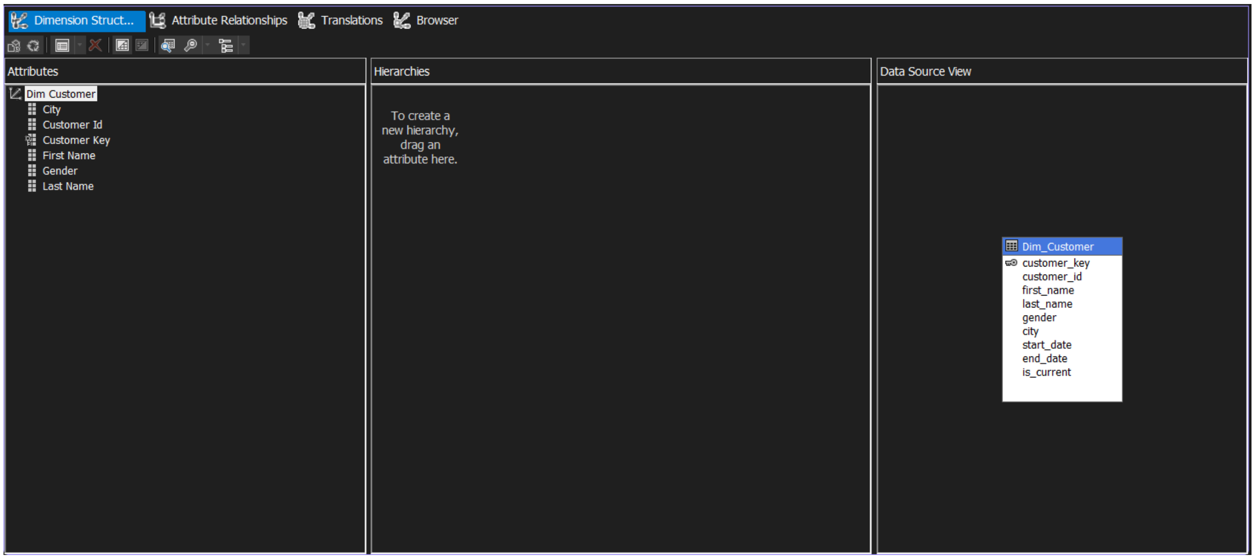

Dimension Wizard
Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.
Available attributes:
Attribute Name
Enable Browsing
Attribute Type
Seat Key
Regular
Seat Id
Regular
Hall Id
Regular
Seat Row
Regular
Seat Number
Regular
Seat Type
Regular
< Back
Next >
Finish >>|
Cancel

1. **Dim_Date dimension -** Date dimension created from Dim_Date.

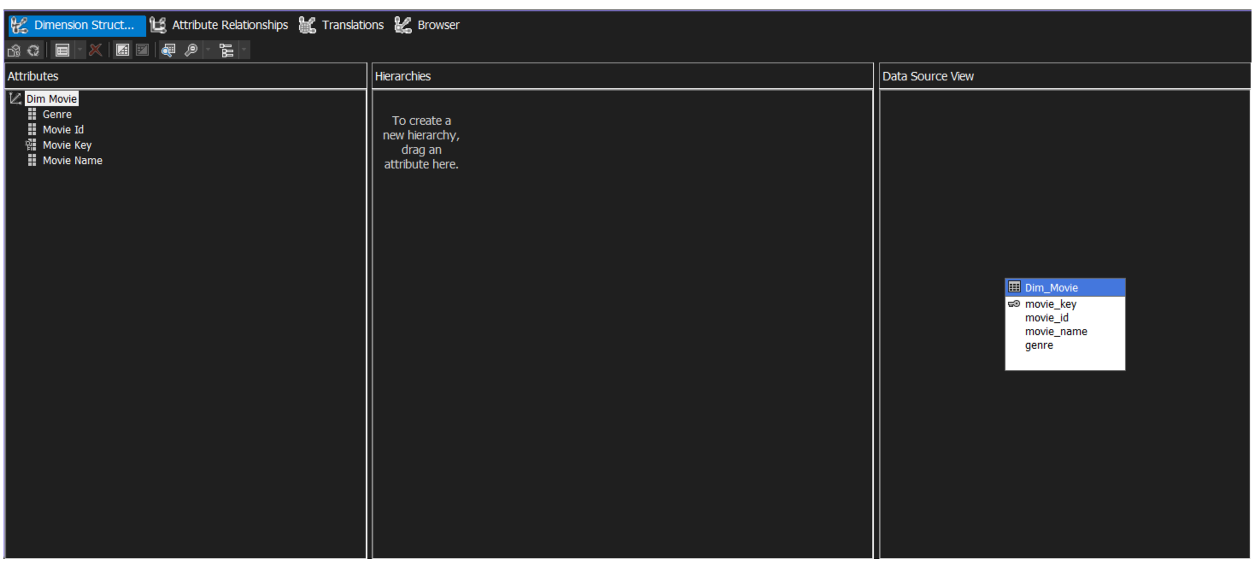




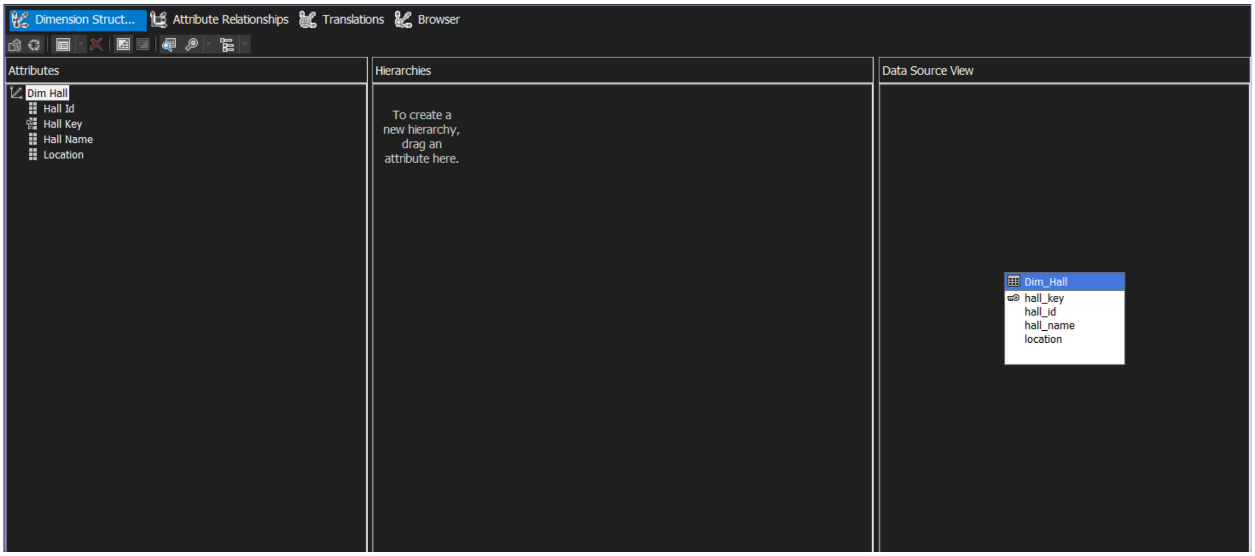
2. **Dim_Customer dimension-** Customer dimension created from Dim_Customer.



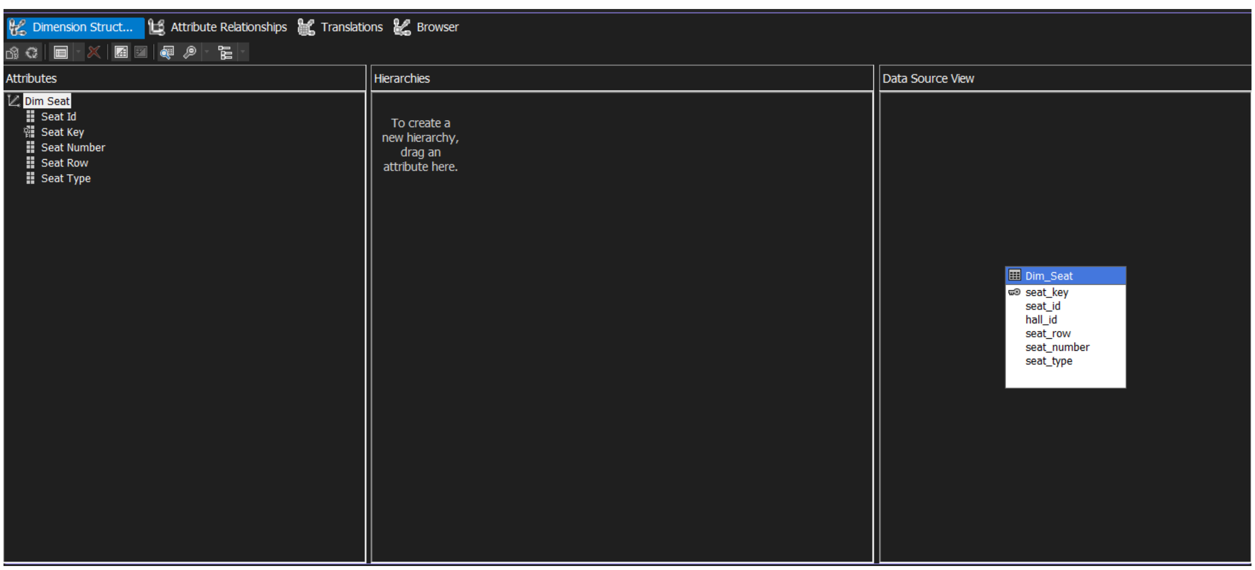
3. **Dim_Movie dimension-** Movie dimension created from Dim_Movie.



4. **Dim_Hall dimension-** Hall dimension created from Dim_Hall.

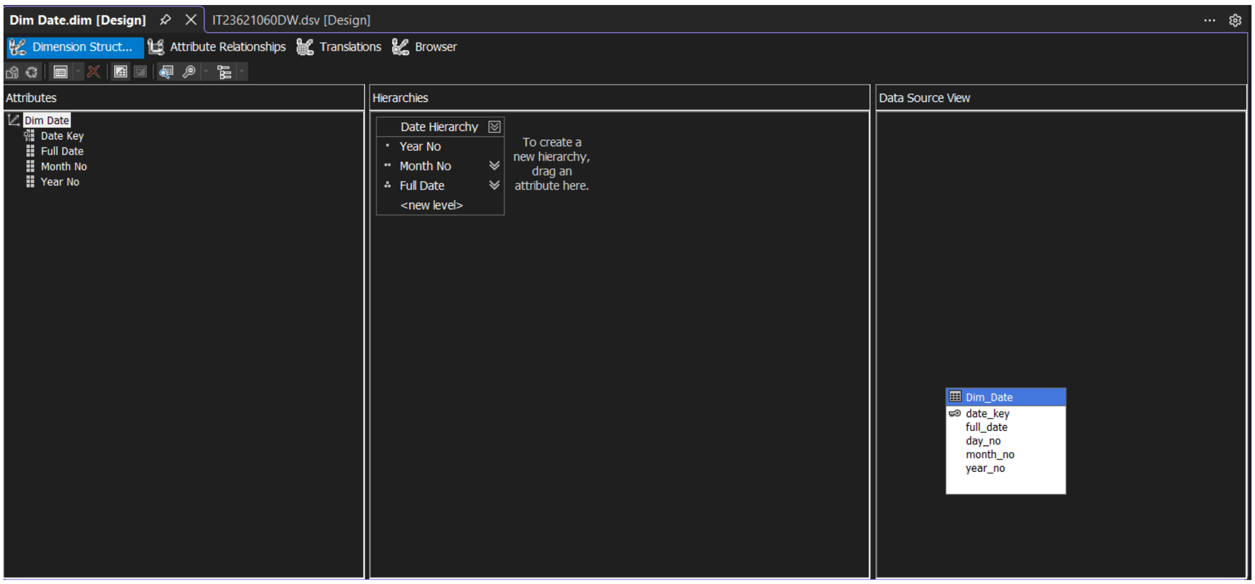


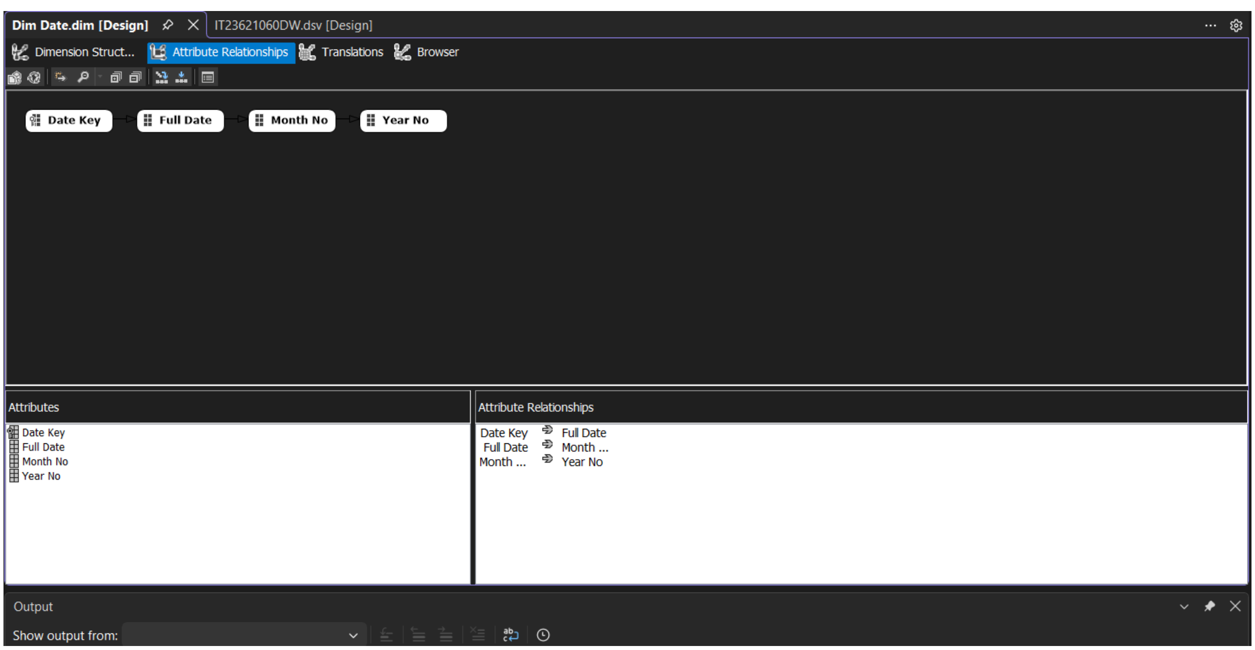
5. **Dim_Seat dimension-** Seat dimension created from Dim_Seat.



2.5. Creating the hierarchy

A date hierarchy was implemented in the Dim_Date dimension using Year No, Month No, and Full Date. This hierarchy supports hierarchical browsing and drill-down analysis in the cube.

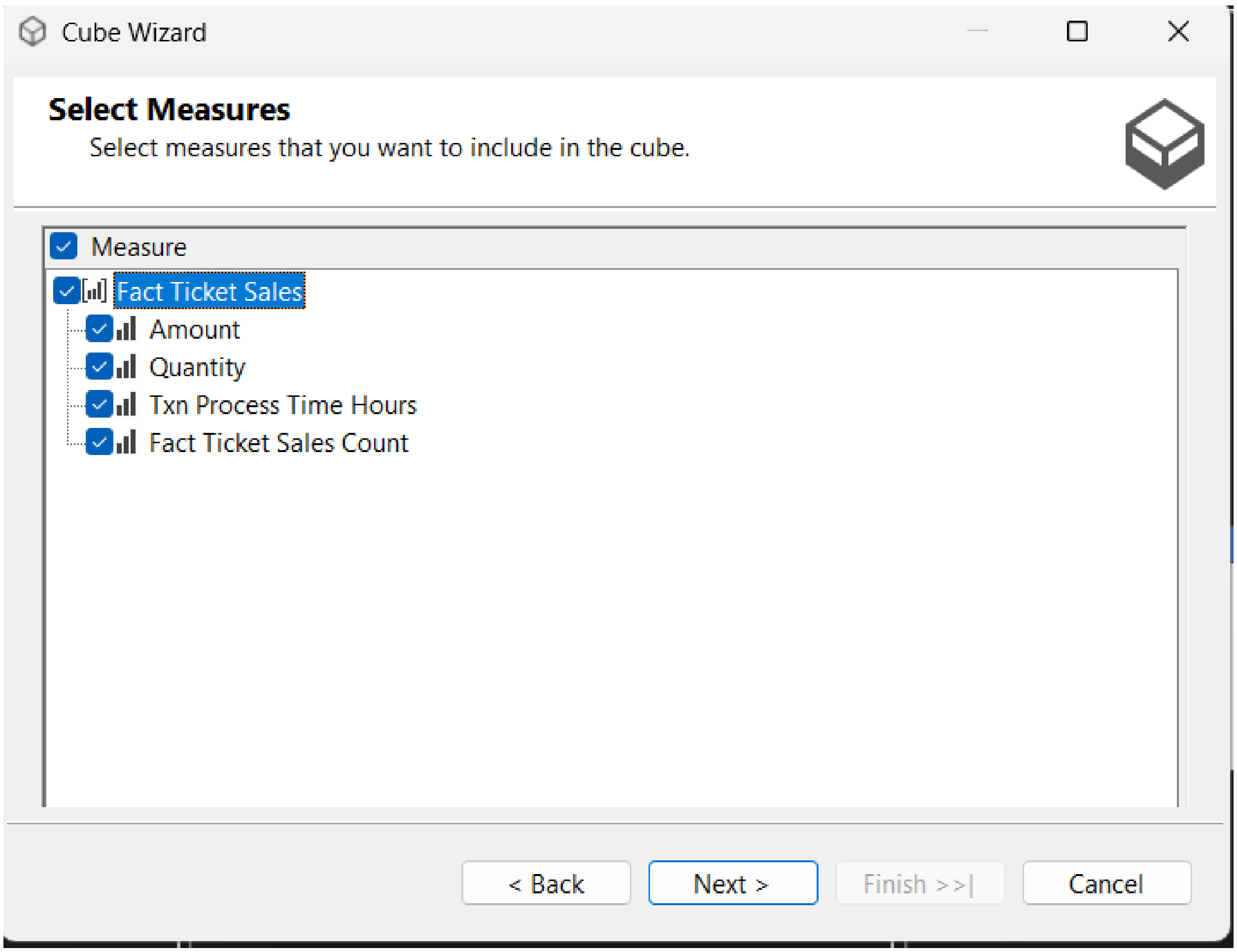


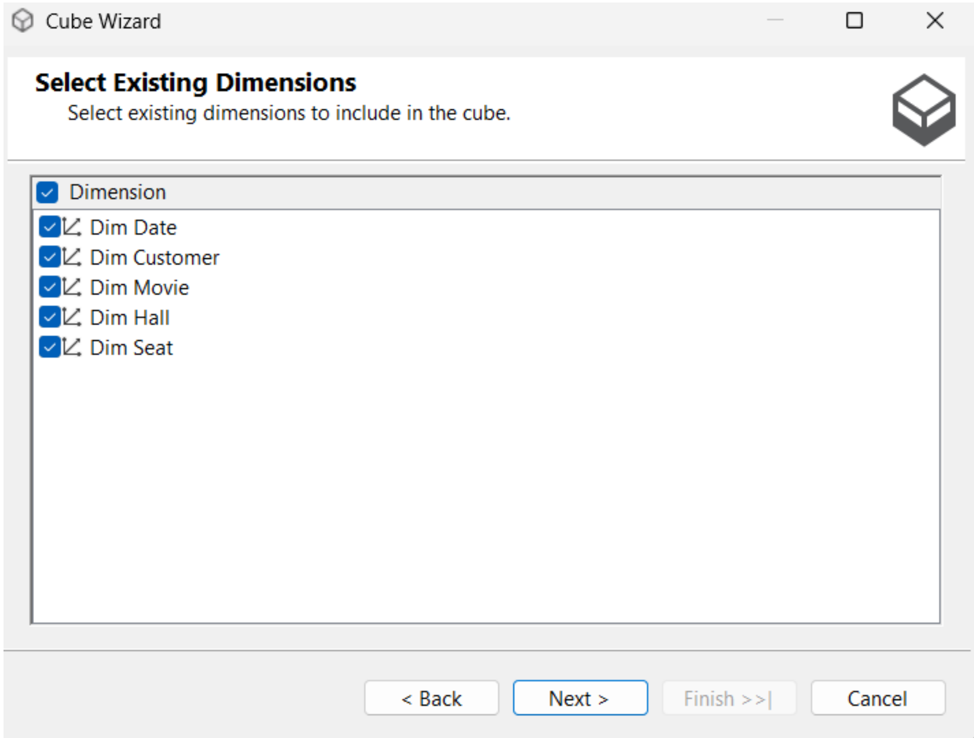


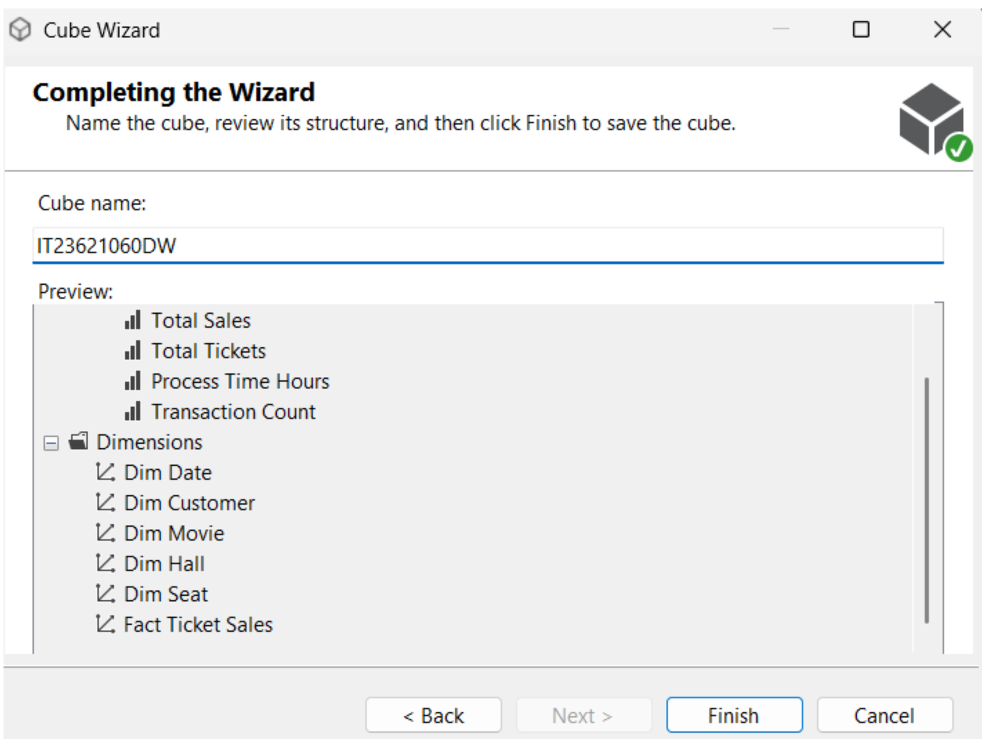
Date hierarchy created in the Date dimension.

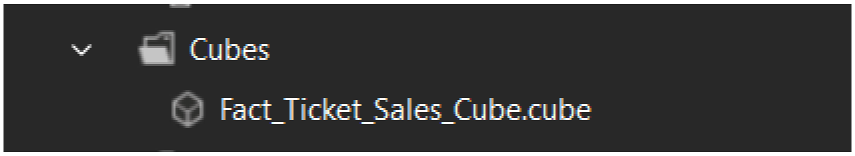
2.6. Creating the cube

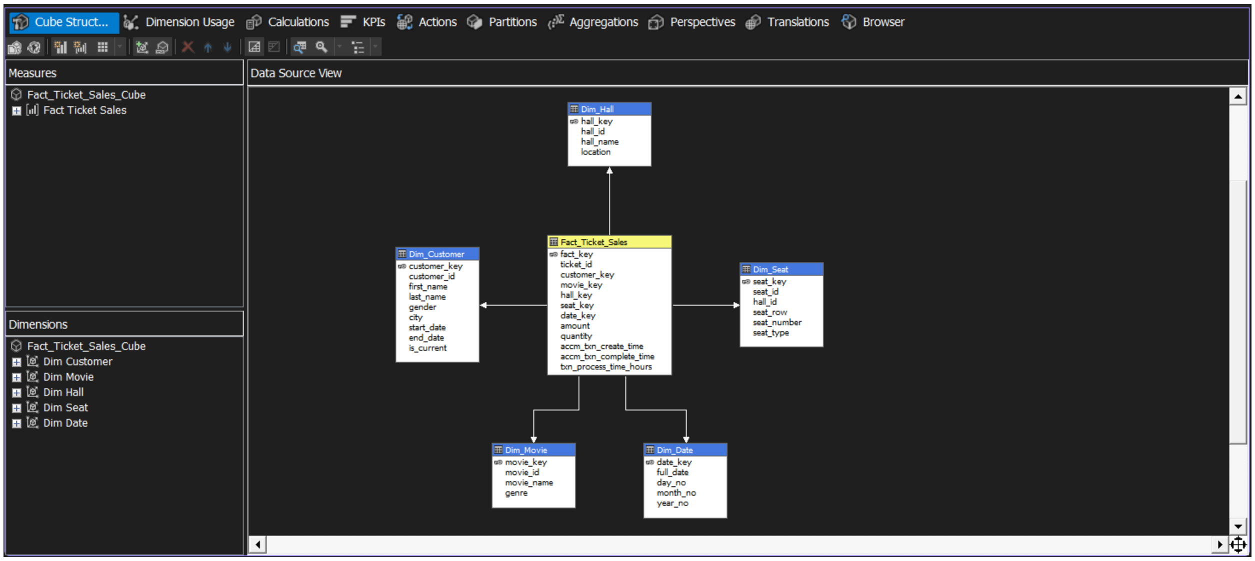
The cube was created using Fact_Ticket_Sales as the measure group. Measures such as Total Sales, Total Tickets, Process Time Hours, and Transaction Count were selected. The dimensions Customer, Movie, Hall, Seat, and Date were added to the cube.




Cube Wizard
Select Existing Dimensions
Select existing dimensions to include in the cube.
Dimension
Dim Date
Dim Customer
Dim Movie
Dim Hall
Dim Seat
< Back
Next >
Finish >>|
Cancel


Cube Wizard
Completing the Wizard
Name the cube, review its structure, and then click Finish to save the cube.
Cube name:
IT23621060DW
Preview:
Total Sales
Total Tickets
Process Time Hours
Transaction Count
Dimensions
Dim Date
Dim Customer
Dim Movie
Dim Hall
Dim Seat
Fact Ticket Sales
< Back
Next >
Finish
Cancel



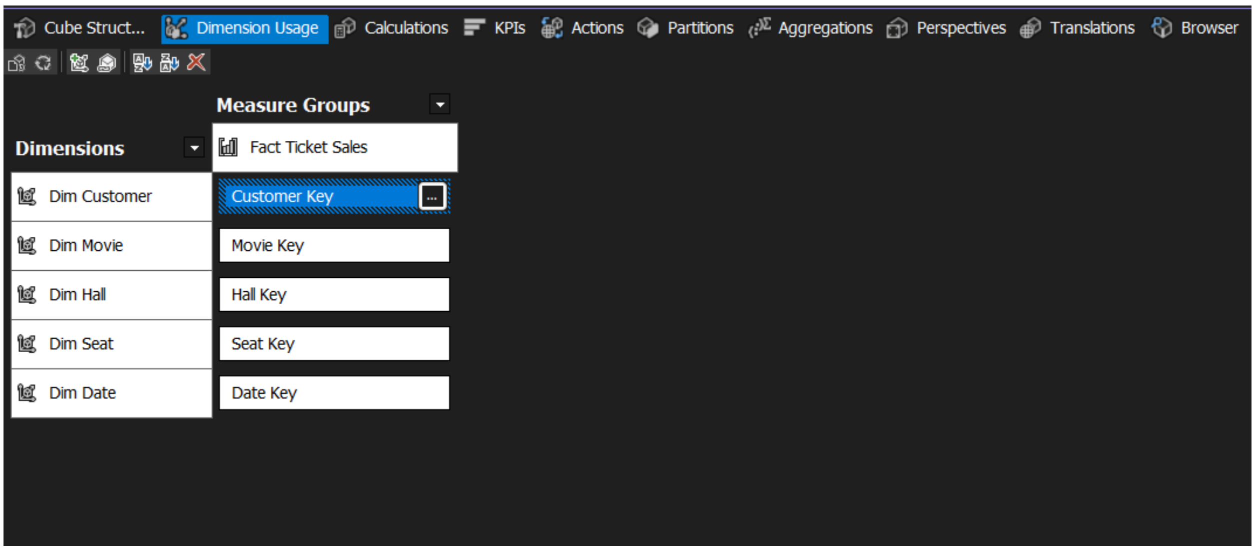


Cube created using Fact_Ticket_Sales as the measure group.

The cube was designed using Fact_Ticket_Sales as the central measure group because it contains the numeric measures required for analysis. The dimensions were added to support slicing and drilling across customer, movie, hall, seat, and date perspectives.

2.7. Dimension usage

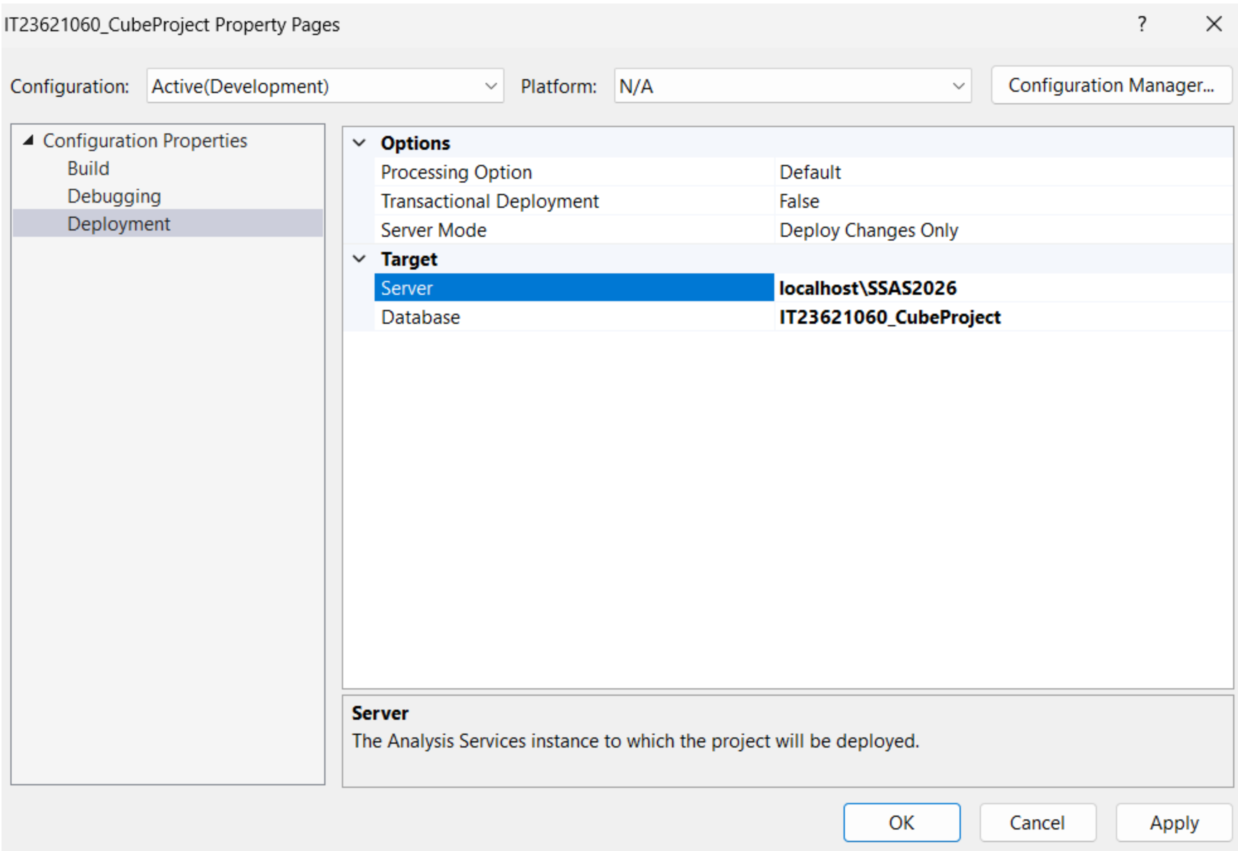
The Dimension Usage tab was checked to ensure that all dimensions were correctly mapped to the fact table using their surrogate keys. This confirmed the logical relationship between the measure group and each dimension.



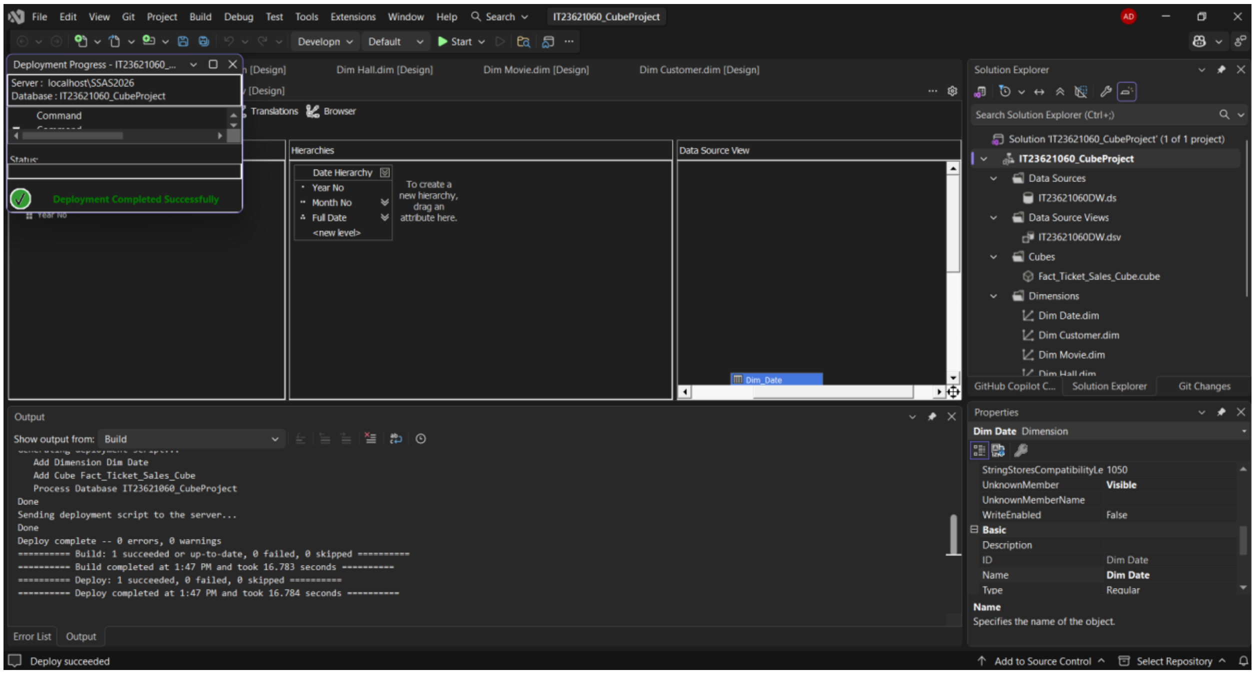
Dimension Usage showing correct relationships between fact and dimensions.

2.8. Deployment and processing

The SSAS project was deployed to the local Analysis Services instance localhost\SSAS2026. After deployment, the database was processed successfully so that the cube could be browsed.



Deployment target set to the local SSAS instance.



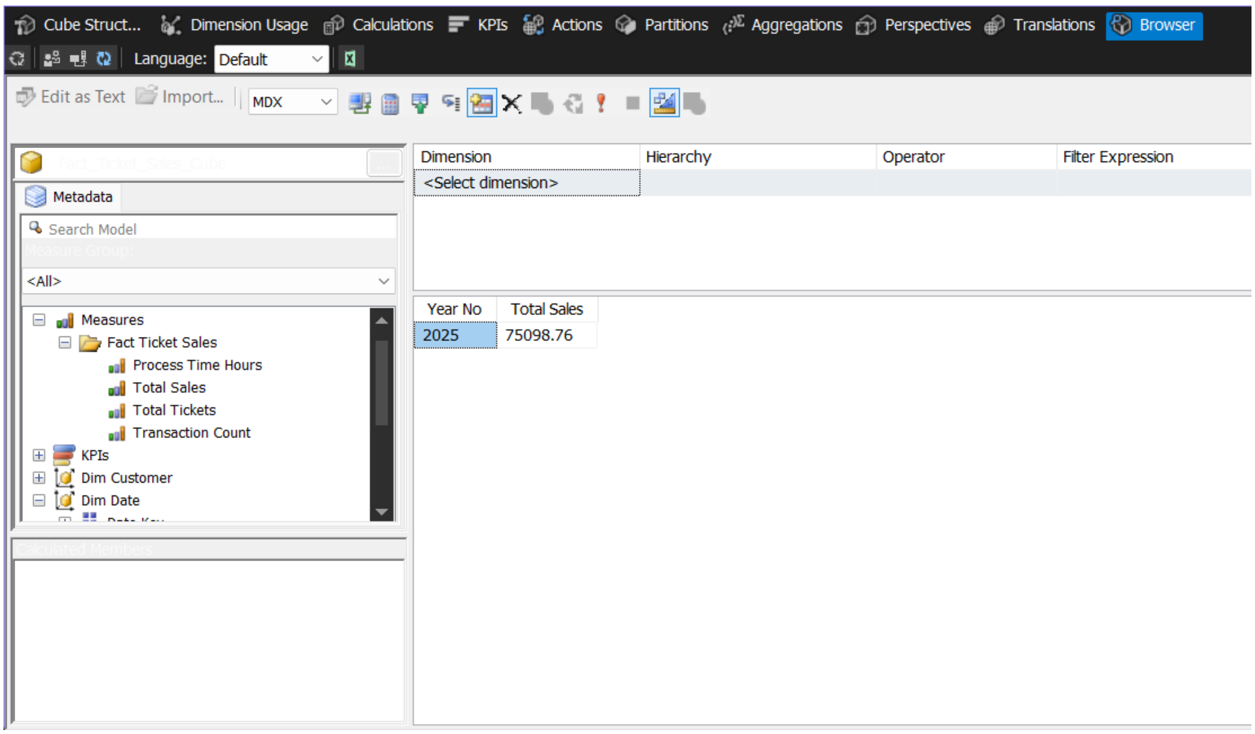
Successful deployment and processing of the cube.

Deployment published the cube structure to the Analysis Services server, while processing loaded the warehouse data into the cube so that it could be queried from the Browser tab and external tools such as Excel.

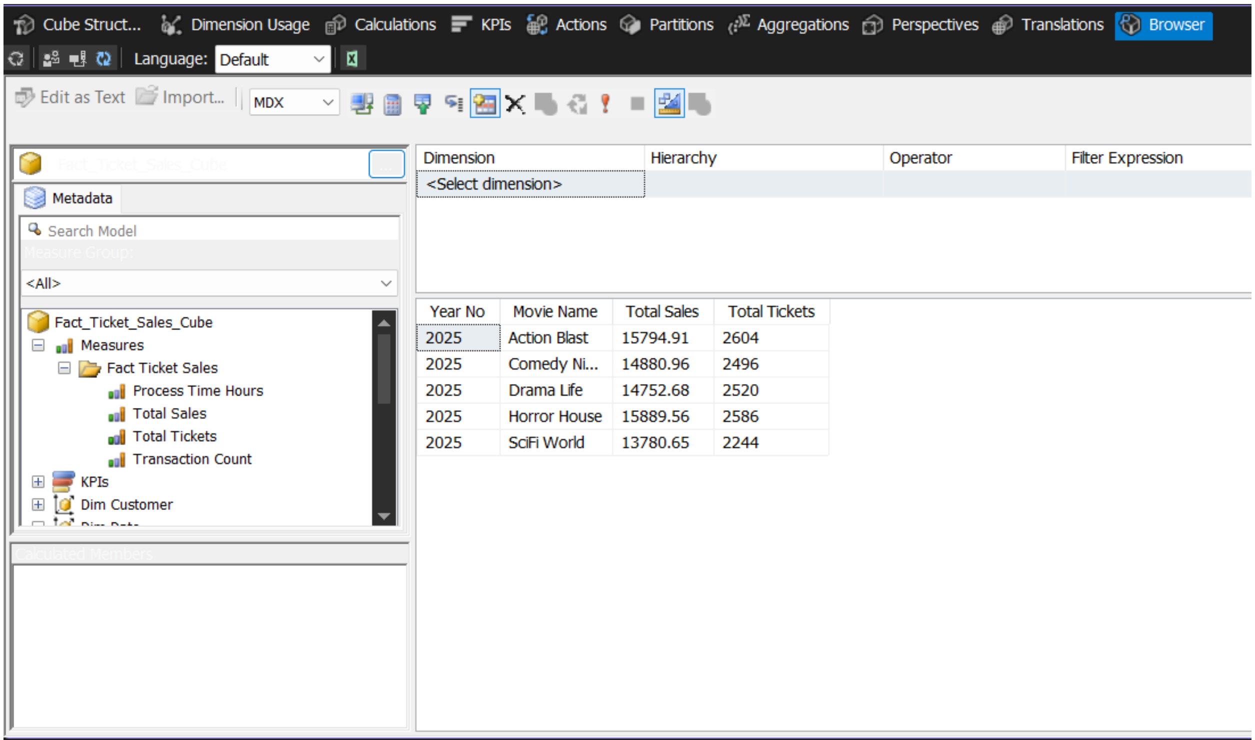
2.9. Browsing the cube

The Browser tab was used to validate the cube. Measures and dimensions were dragged into the query area to confirm that the cube returned correct analytical results.

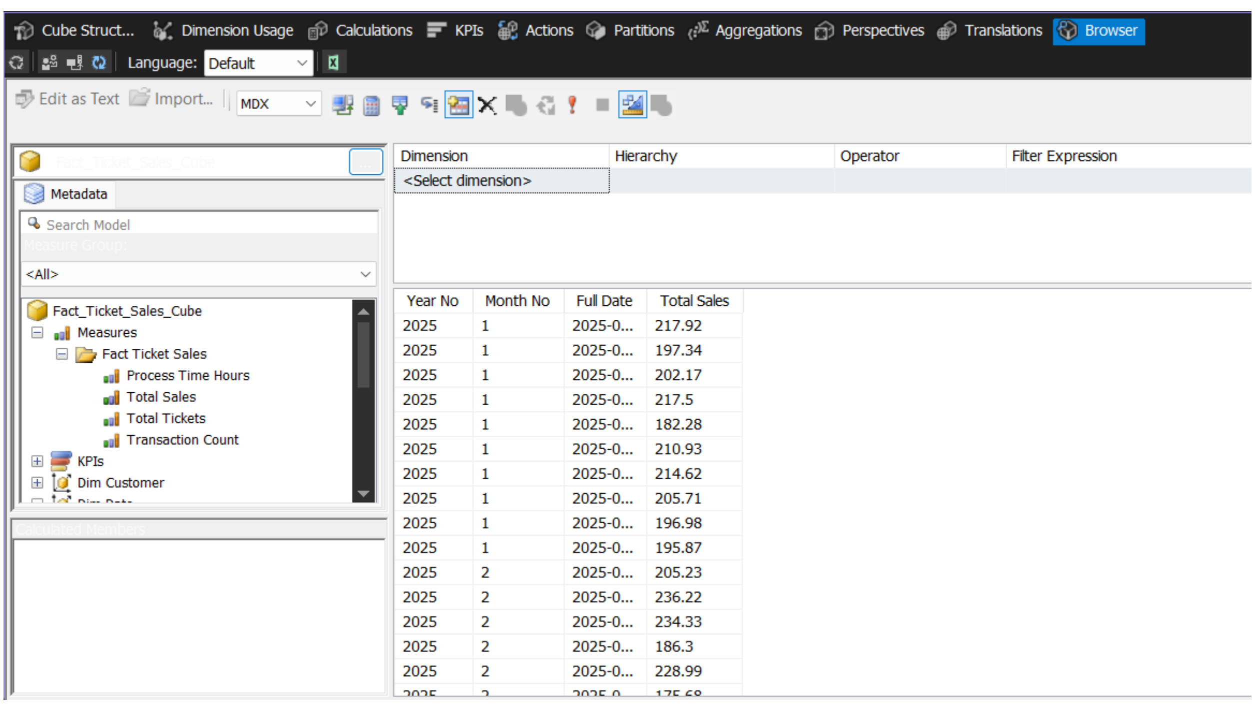
These browser results confirmed that the measures, dimensions, and hierarchy were functioning correctly in the deployed cube.



Cube browser showing Total Sales by Year.



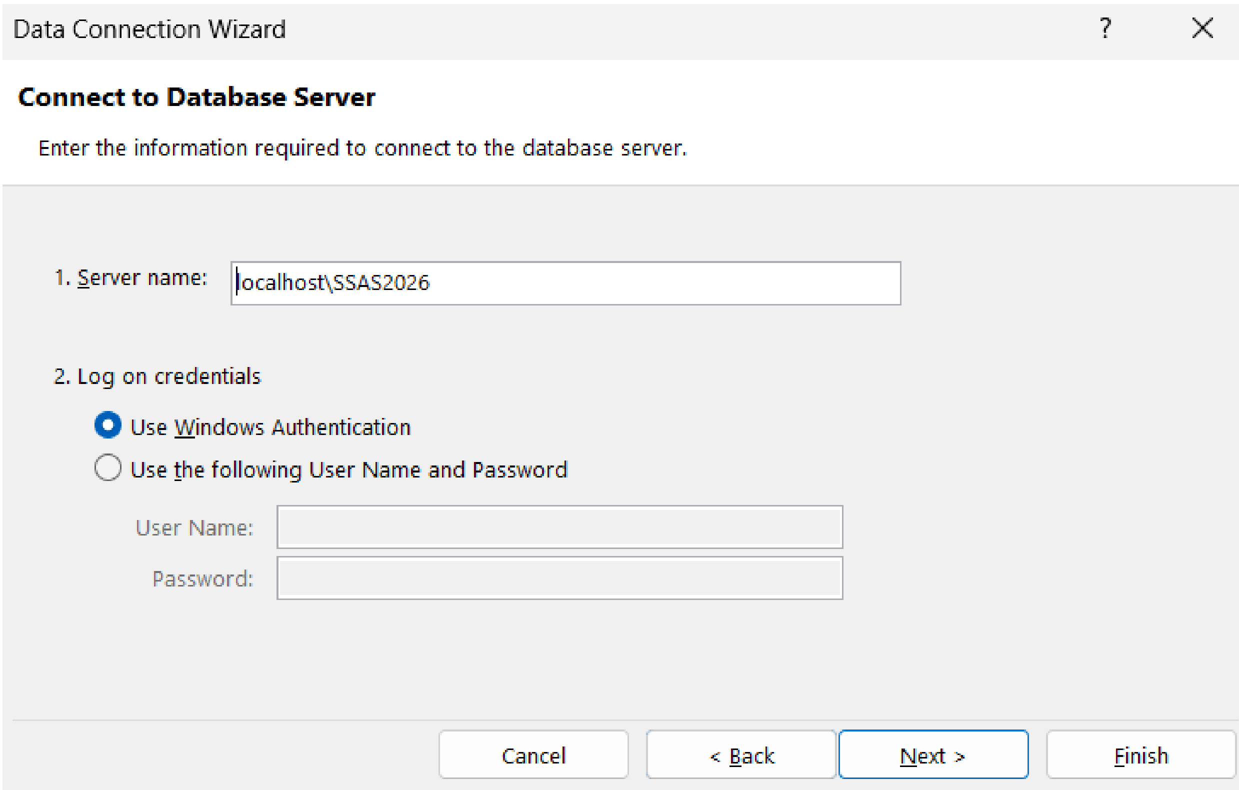
Cube browser showing Total Sales and Total Tickets by Movie.

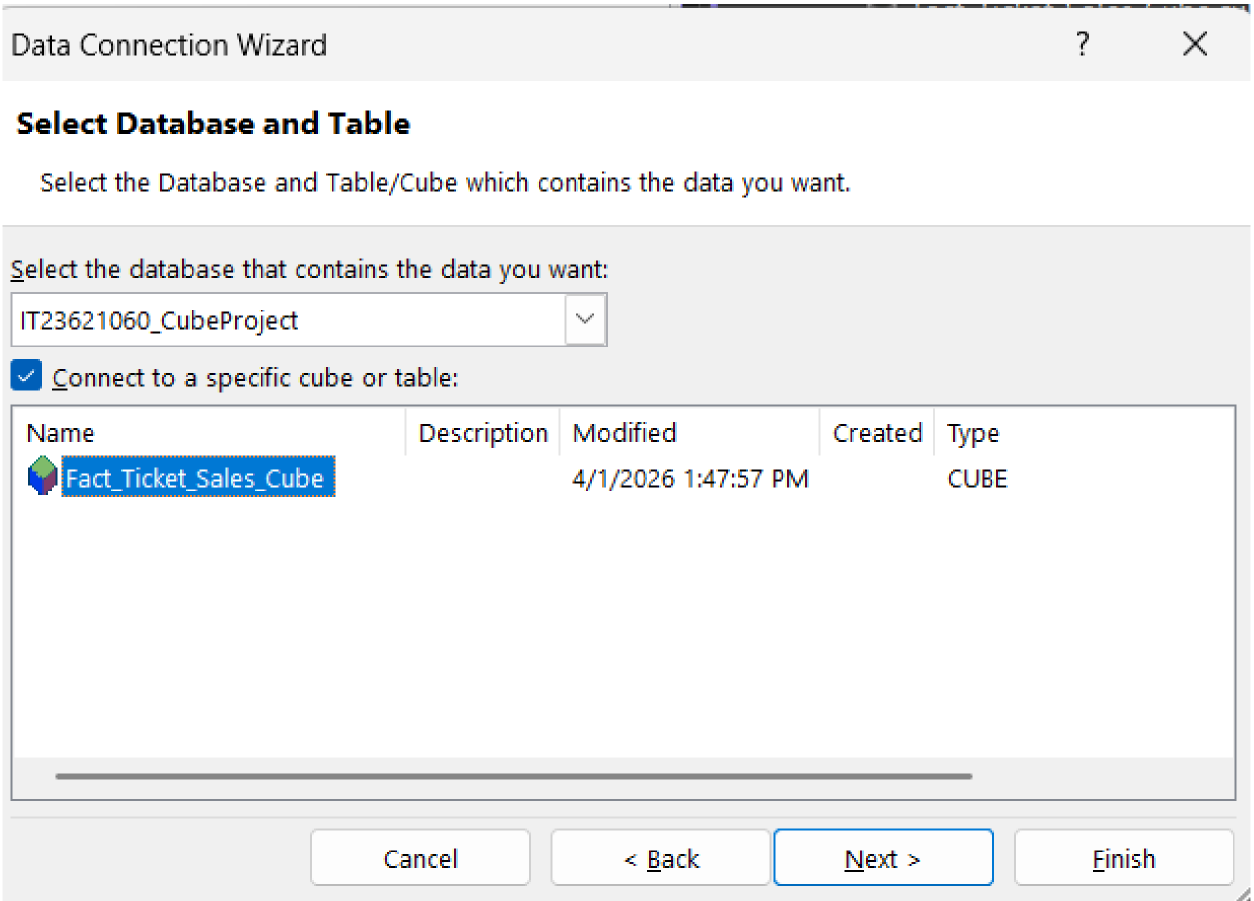


Cube browser showing Date Hierarchy with Total Sales.

Step 3 — Excel OLAP operations

Excel was connected to the deployed SSAS cube using the Analysis Services connection option. A PivotTable was created in a new worksheet for OLAP analysis.

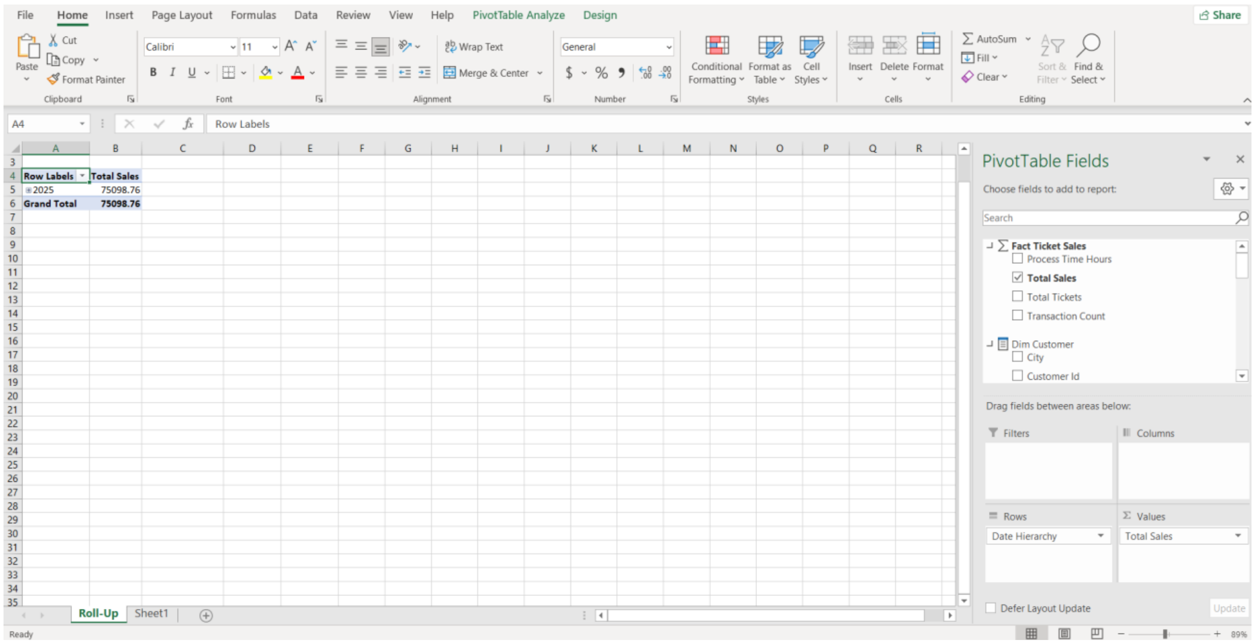




Excel connected to the deployed SSAS cube using an Analysis Services connection.

3.1. Roll-up

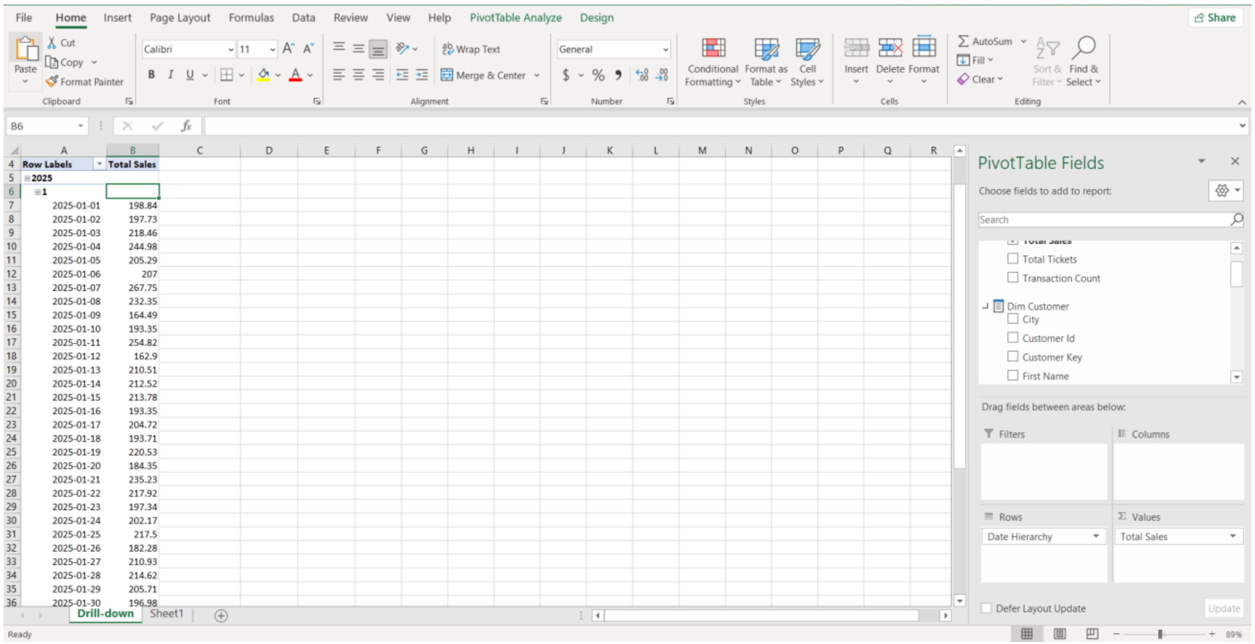
Roll-up was demonstrated by summarizing Total Sales at the Year level using a PivotTable.



Roll-up operation showing Total Sales by Years

3.2. Drill-down

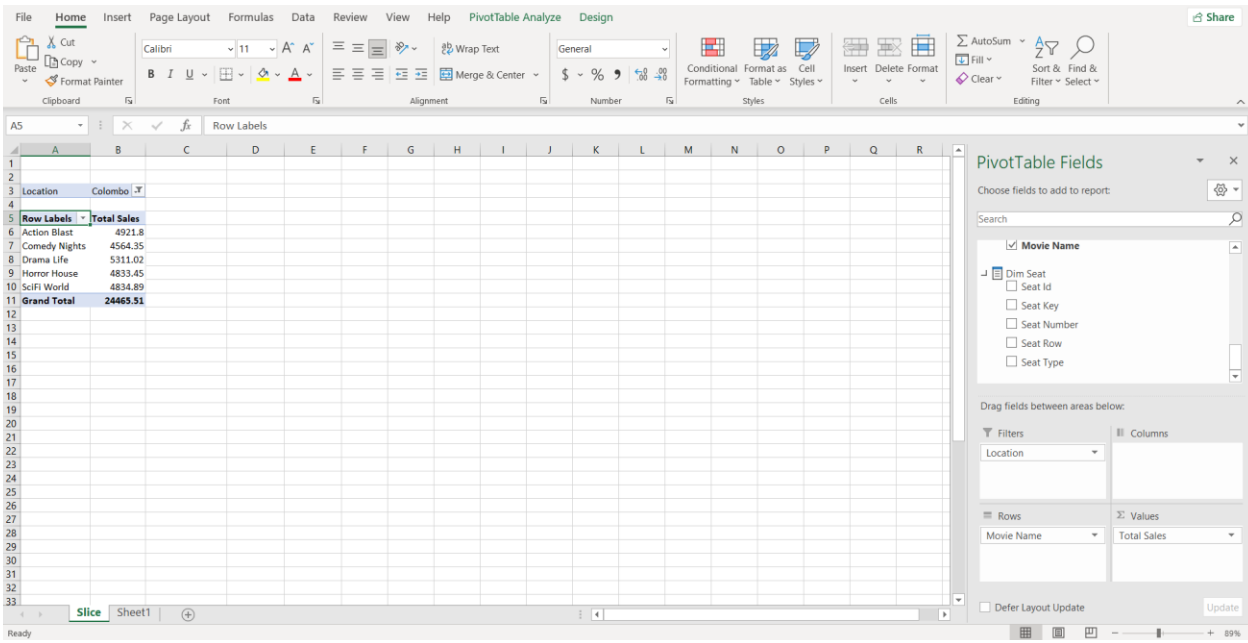
Drill-down was demonstrated using the Date Hierarchy, allowing navigation from Year level to Month and Full Date levels.



Drill-down operation using the Date Hierarchy.

3.3. Slice

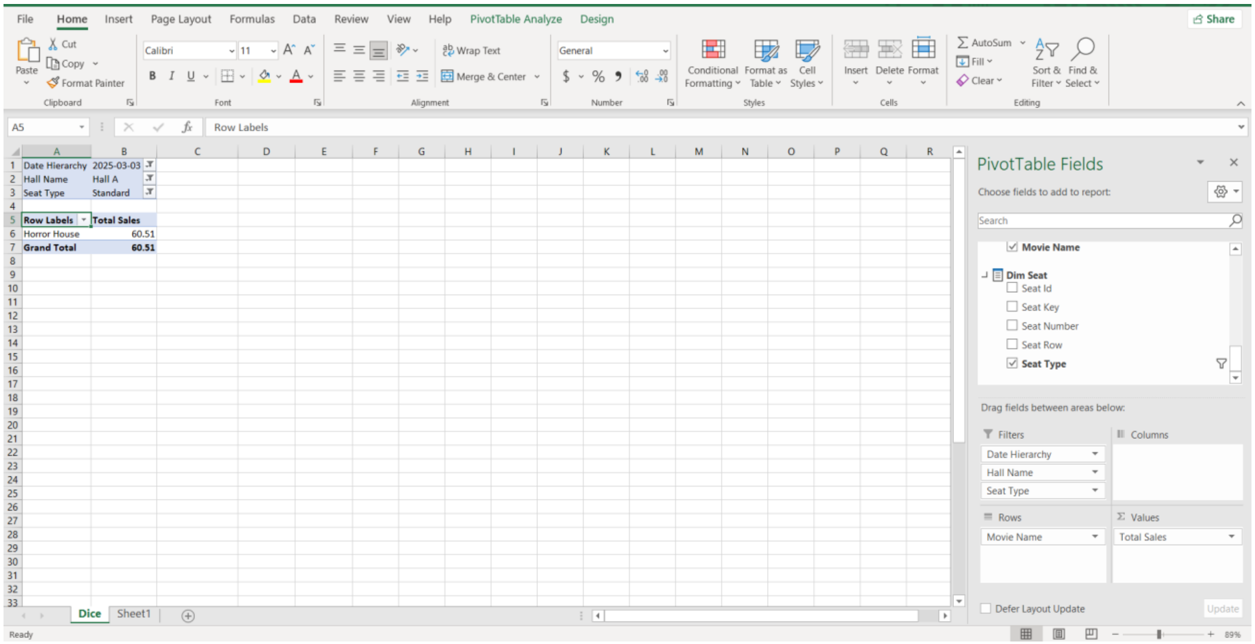
Slice was demonstrated by applying a filter on a single dimension value, such as one hall or one location, to analyze a smaller subset of the cube.



Slice operation using a single filter value.

3.4. Dice

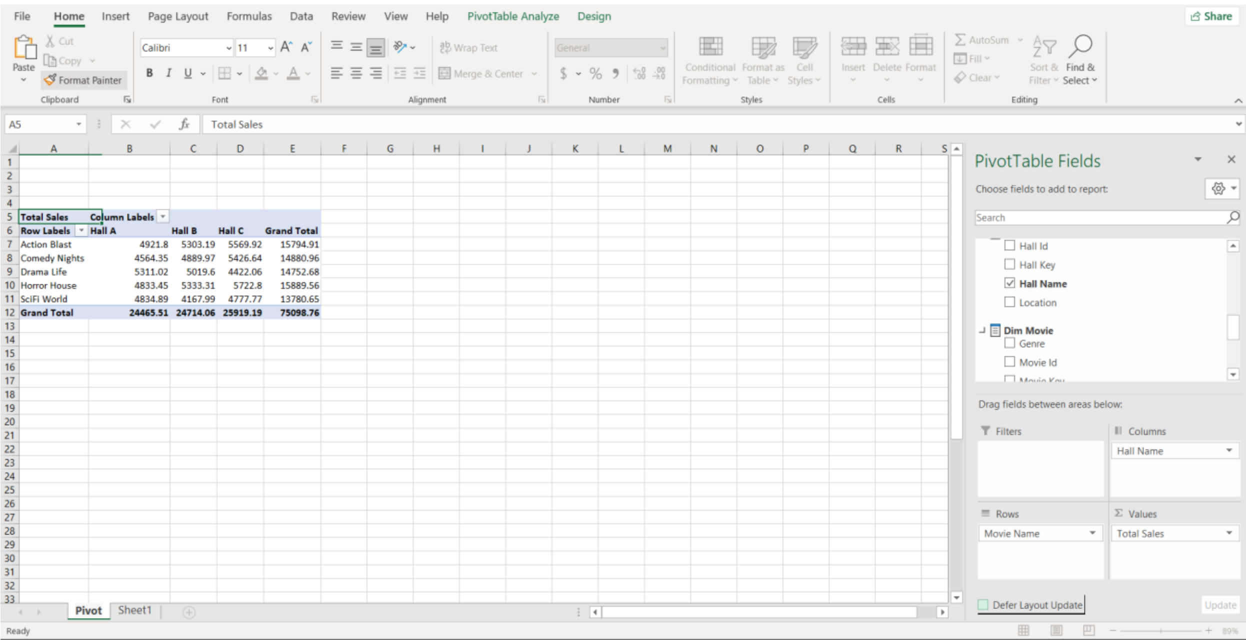
Dice was demonstrated by applying multiple filters simultaneously, such as Year and Hall, to analyze a more specific subset of the data.

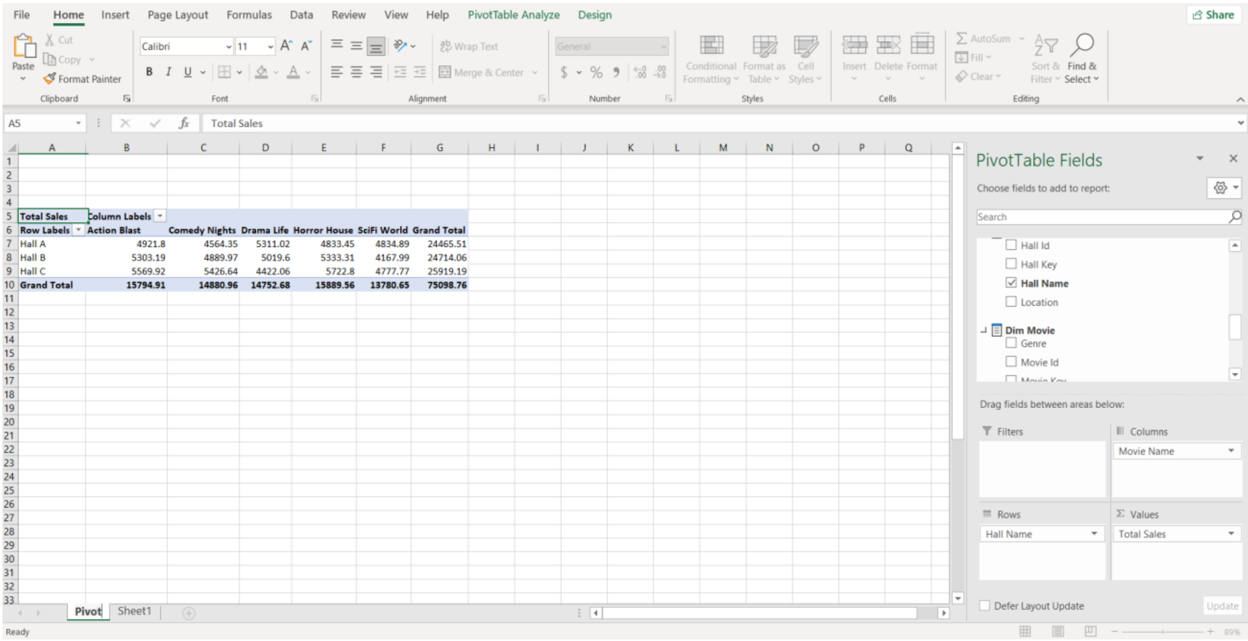


Dice operation using multiple filter values.

3.5. Pivot

Pivot was demonstrated by arranging Movie Name in rows and Hall Name in columns with Total Sales as the value, then switching the arrangement to view the same data from a different perspective





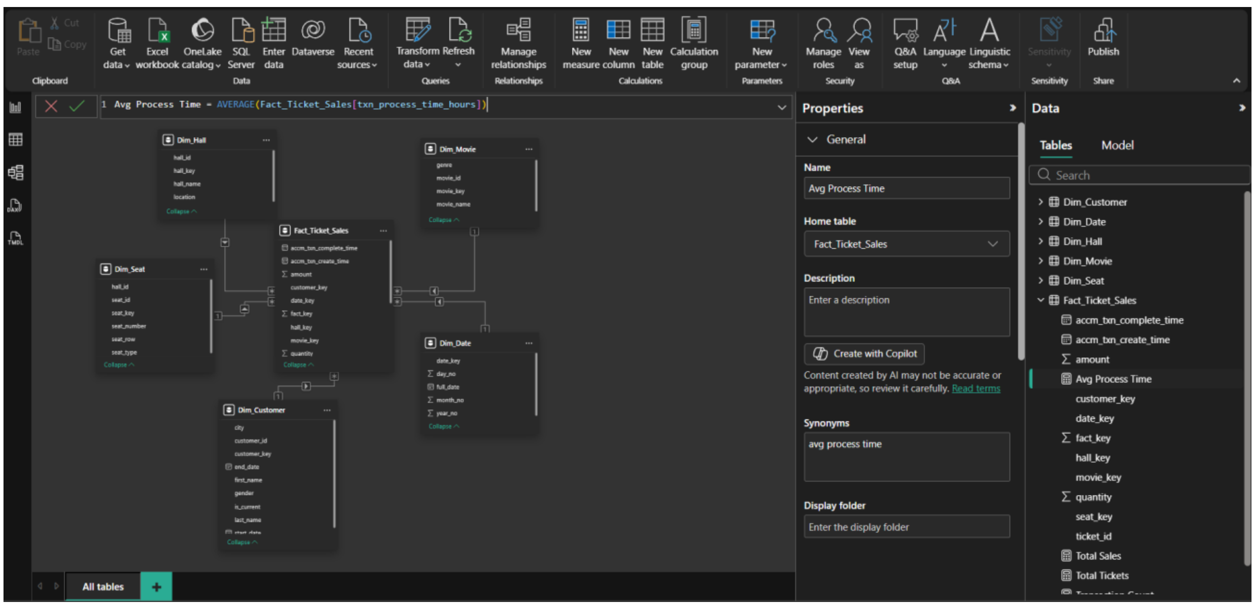
Pivot operation showing Total Sales by Movie and Hall

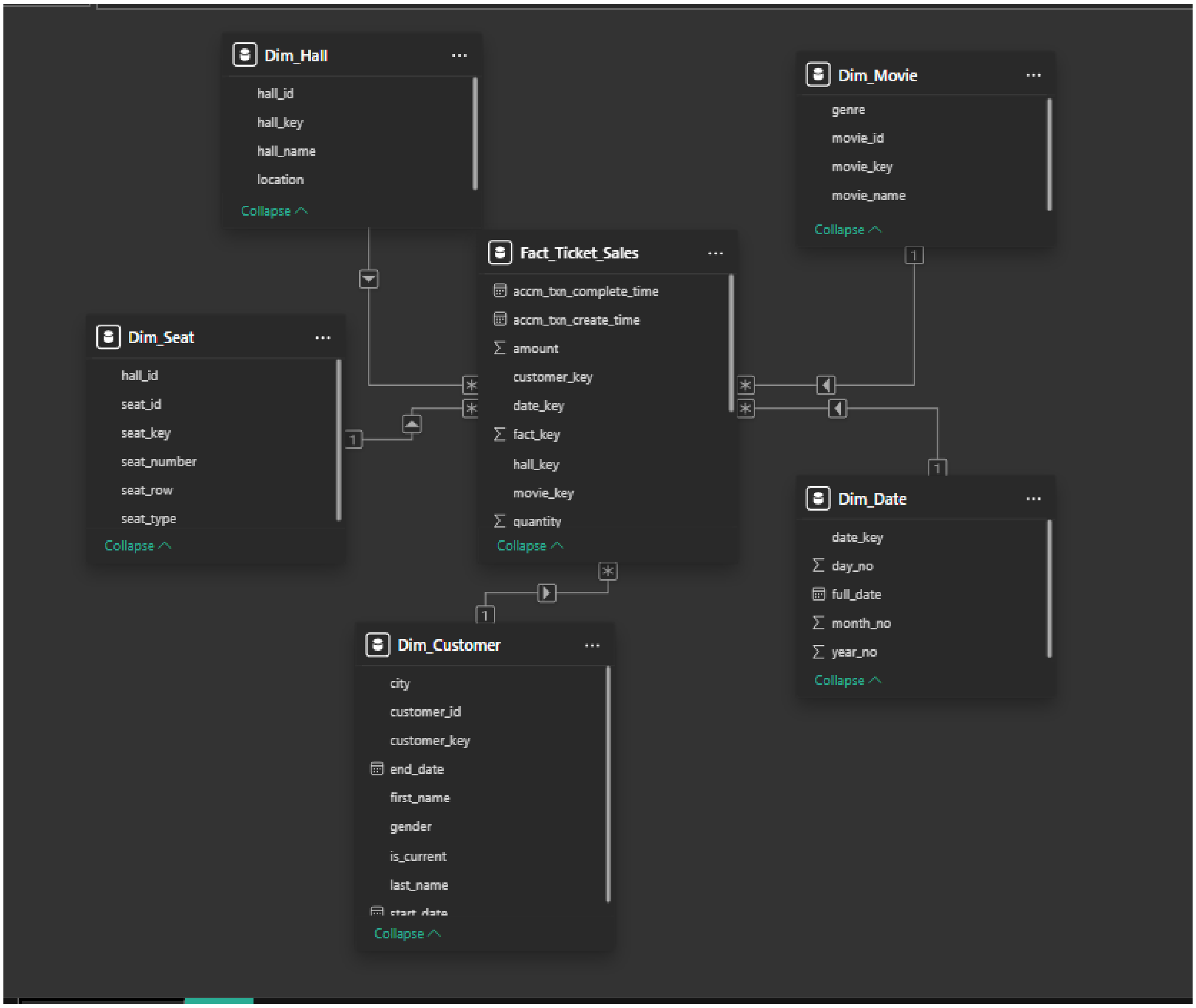
These PivotTable views confirmed that the cube could successfully support all required OLAP operations in Excel.

Step 4 — Power BI reports

4.1. Loading the warehouse into Power BI

Power BI Desktop was connected to the IT23621060DW database using SQL Server Import mode. After loading the fact and dimension tables, the model relationships were verified and DAX measures were created for reporting. The following report pages were then developed according to the assignment requirements.





Power BI data model with fact and dimension relationships.

4.2. Measures created in Power BI

The following DAX measures were created for reporting: Total Sales, Total Tickets, Transaction Count, and Average Process Time.

```
Total Sales = SUM(Fact_Ticket_Sales[amount])
```

```
Total Tickets = SUM(Fact_Ticket_Sales[quantity])
```

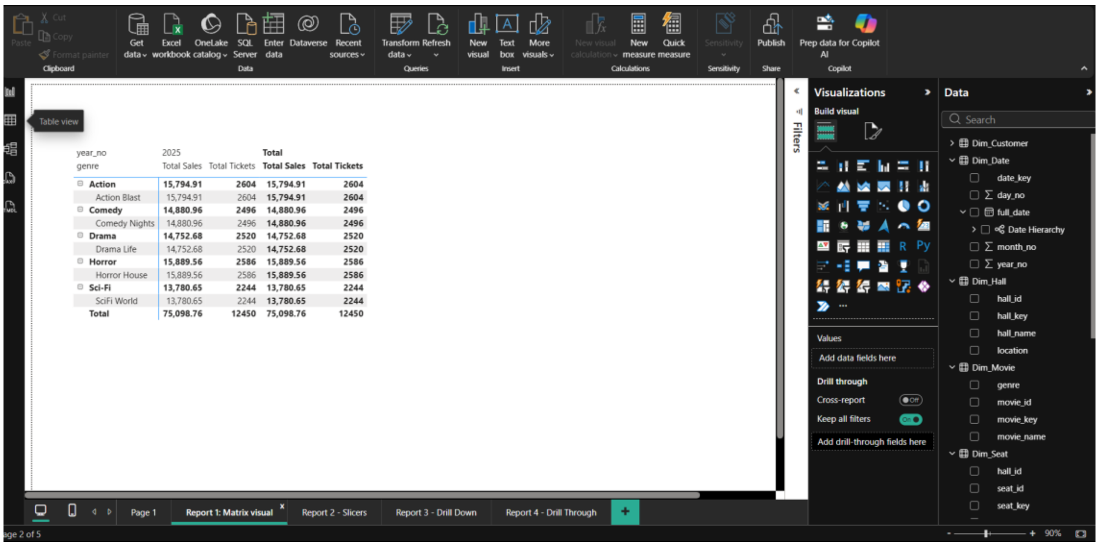
```
Transaction Count = COUNT(Fact_Ticket_Sales[fact_key])
```

```
Avg Process Time = AVERAGE(Fact_Ticket_Sales[txn_process_time_hours])
```

DAX measures created in Power BI

4.3. Report 1 – Matrix

A Matrix visual was created to display detailed tabular data with row and column groupings. Movie Genre and Movie Name were used in rows, while Year and Month were used in columns. Total Sales and Total Tickets were used as values.



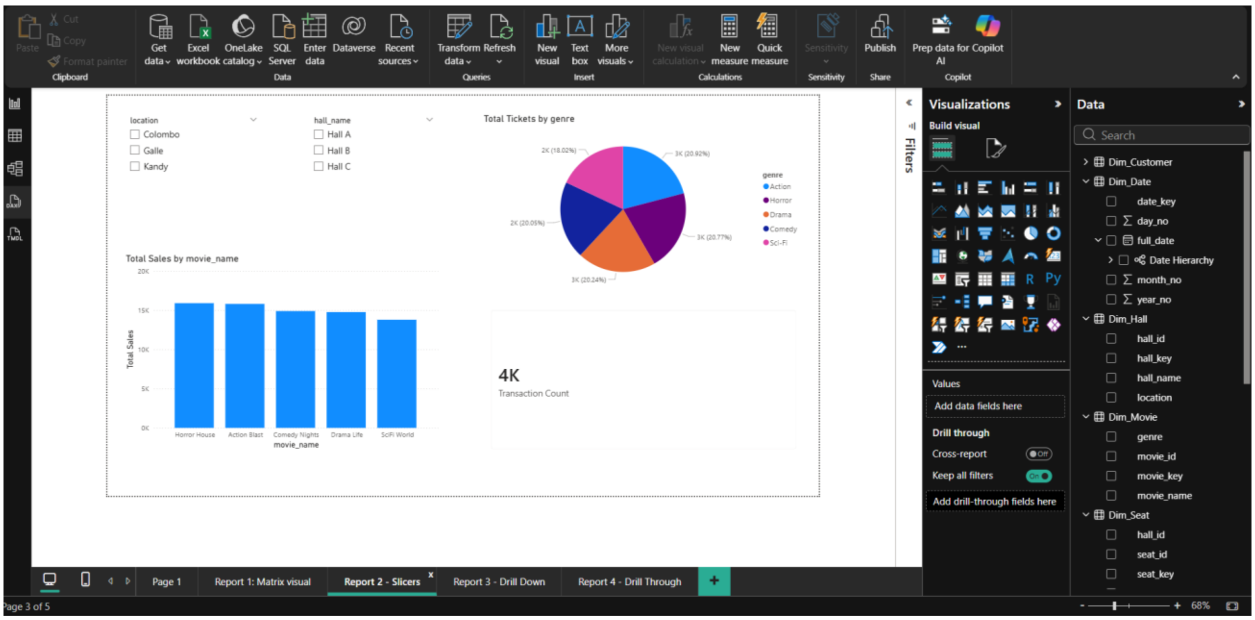
year_no	2025		Total	
genre	Total Sales	Total Tickets	**Total Sales**	**Total Tickets**
⊟ Action	**15,794.91**	**2604**	**15,794.91**	**2604**
Action Blast	15,794.91	2604	**15,794.91**	**2604**
⊟ Comedy	**14,880.96**	**2496**	**14,880.96**	**2496**
Comedy Nights	14,880.96	2496	**14,880.96**	**2496**
⊟ Drama	**14,752.68**	**2520**	**14,752.68**	**2520**
Drama Life	14,752.68	2520	**14,752.68**	**2520**
⊟ Horror	**15,889.56**	**2586**	**15,889.56**	**2586**
Horror House	15,889.56	2586	**15,889.56**	**2586**
⊟ Sci-Fi	**13,780.65**	**2244**	**13,780.65**	**2244**
SciFi World	13,780.65	2244	**13,780.65**	**2244**
Total	**75,098.76**	**12450**	**75,098.76**	**12450**

Matrix visual displaying grouped movie sales data.

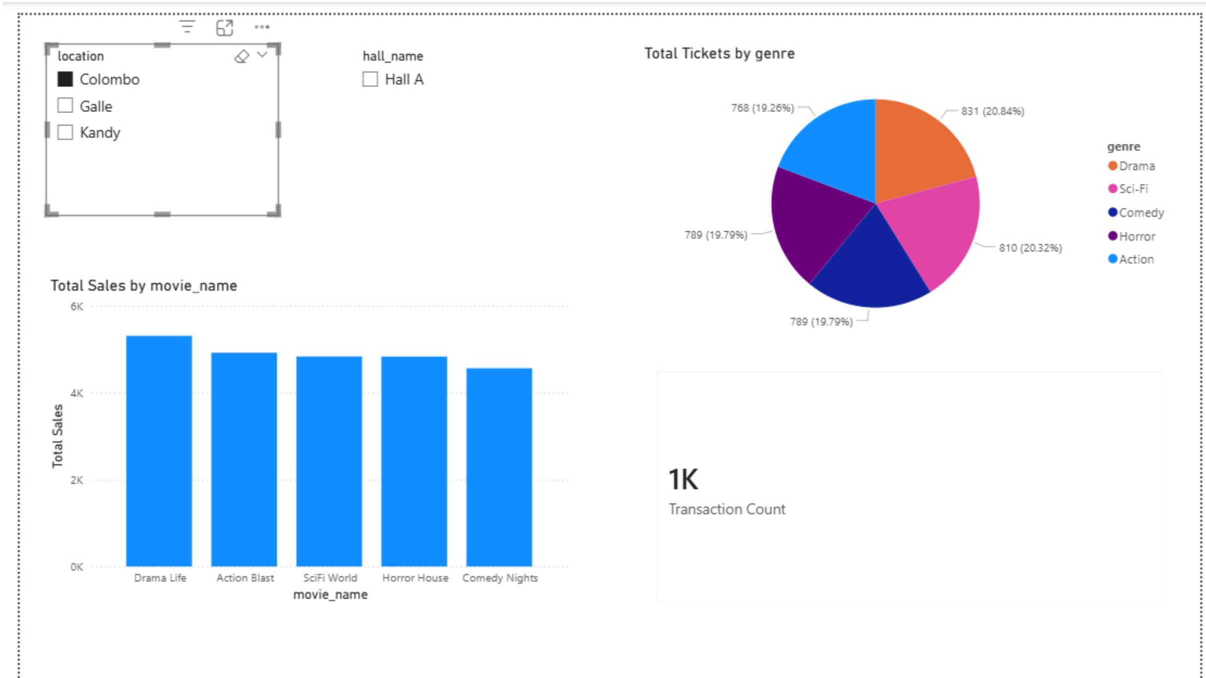
This report provides a tabular view of sales and ticket quantities grouped by movie and time period, making it useful for comparing performance across genres and years.

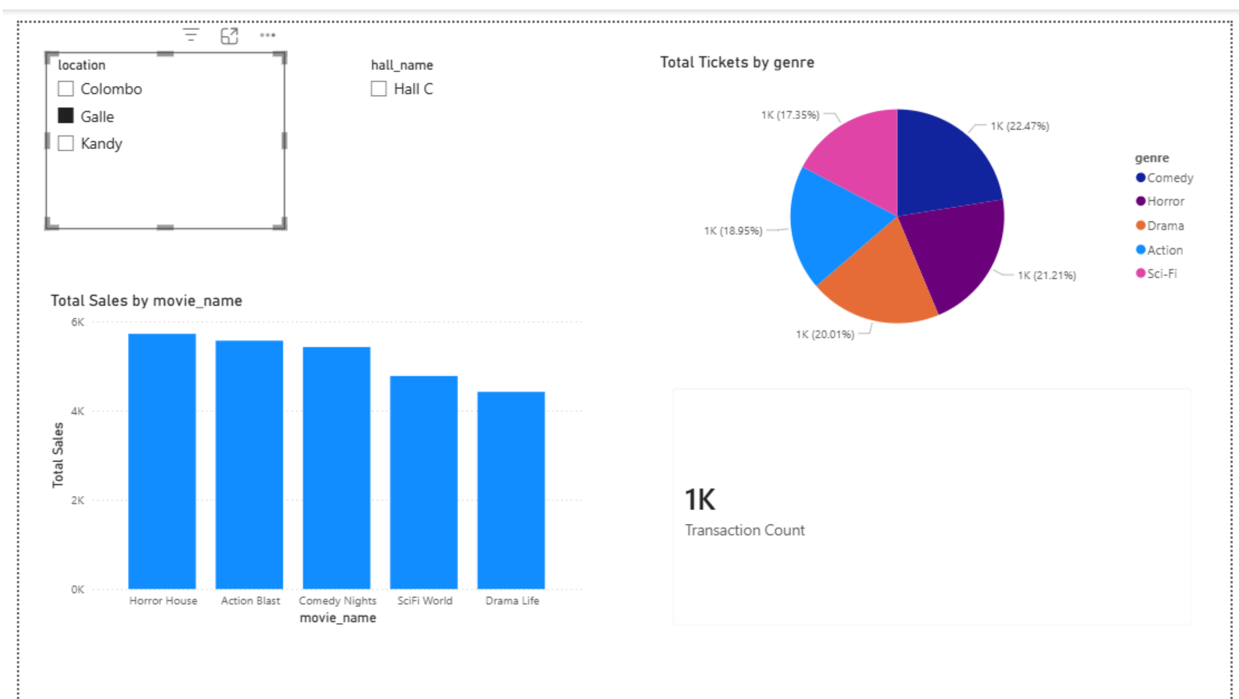
4.4. Report 2 - Multiple slicers with cascading filters

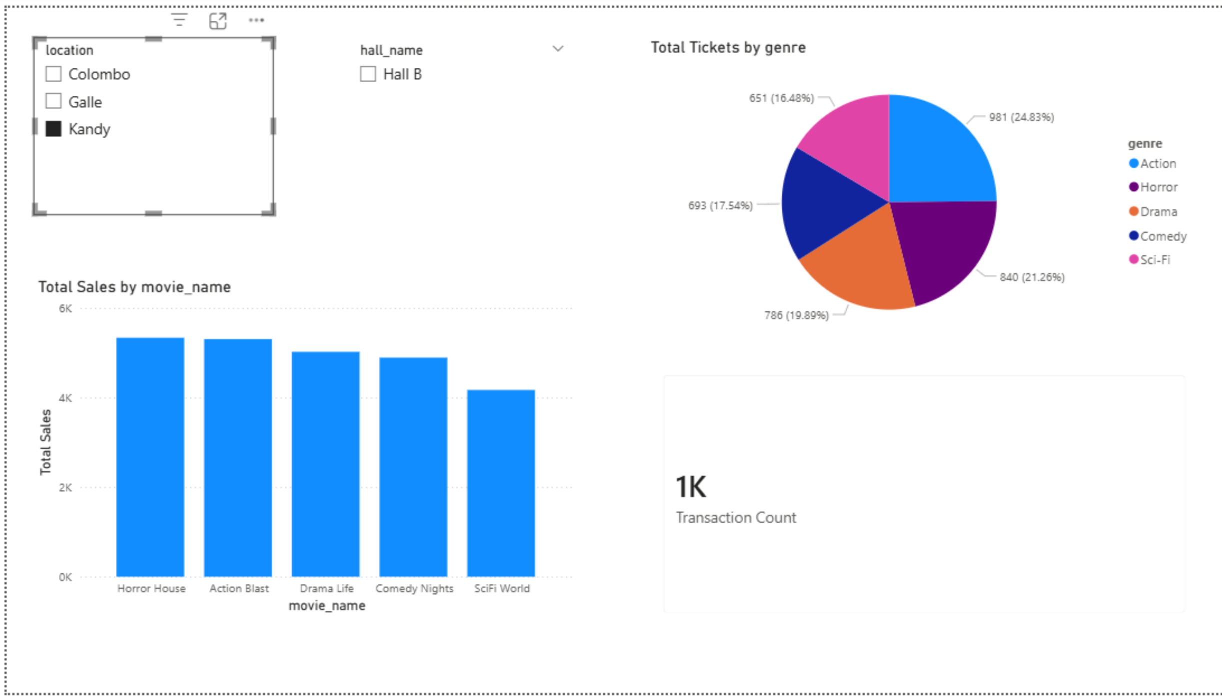
Two slicers were created using location and hall_name. The selection in the first slicer filtered the available values in the second slicer, demonstrating cascading filtering. Additional visuals were added to show sales, ticket distribution, and transaction counts.



Cascading slicers using Location and Hall Name.


location
Colombo
Galle
Kandy
hall_name
Hall A
Total Tickets by genre
768 (19.26%)
831 (20.84%)
789 (19.79%)
810 (20.32%)
789 (19.79%)
genre
Drama
Sci-Fi
Comedy
Horror
Action
Total Sales by movie_name
6K
4K
2K
0K
Total Sales
Drama Life
Action Blast
SciFi World
Horror House
Comedy Nights
movie_name
1K
Transaction Count


location
Colombo
Galle
Kandy
hall_name
Hall C
Total Tickets by genre
1K (17.35%)
1K (22.47%)
1K (18.95%)
1K (21.21%)
1K (20.01%)
genre
Comedy
Horror
Drama
Action
Sci-Fi
Total Sales by movie_name
6K
4K
2K
0K
Total Sales
Horror House
Action Blast
Comedy Nights
SciFi World
Drama Life
movie_name
1K
Transaction Count



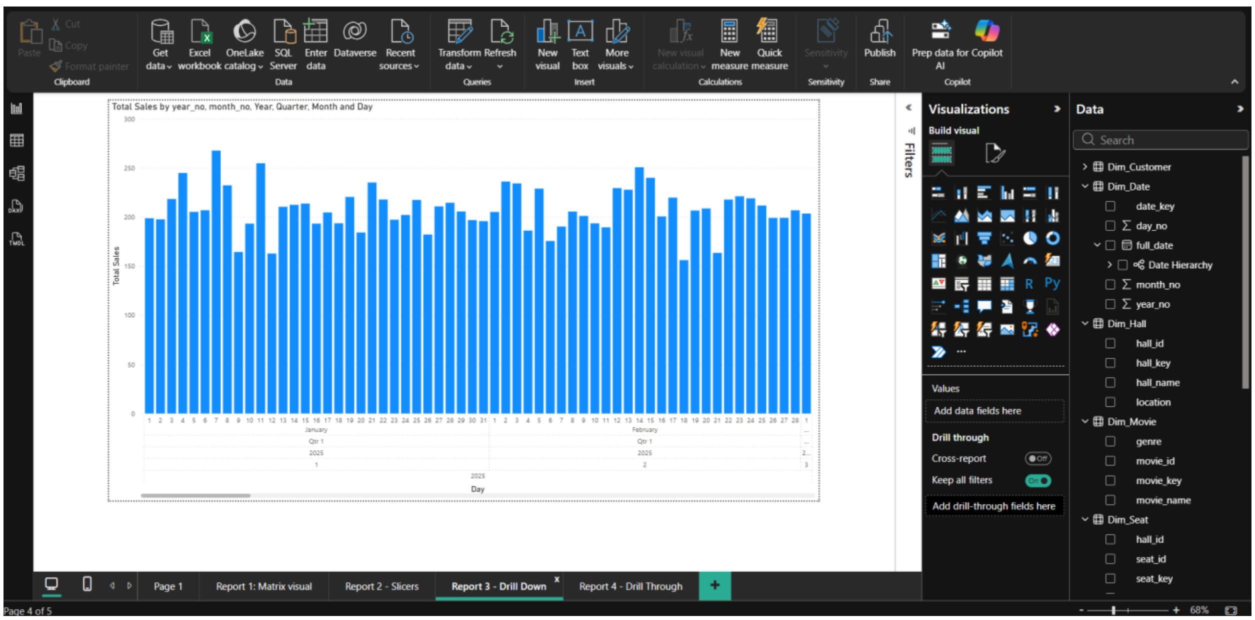
Visuals updated dynamically based on slicer selections.

The first slicer limits the values available in the second slicer, which demonstrates cascading filtering. The report also includes multiple visuals to display changes in sales, tickets, and transaction counts based on the selected filters.

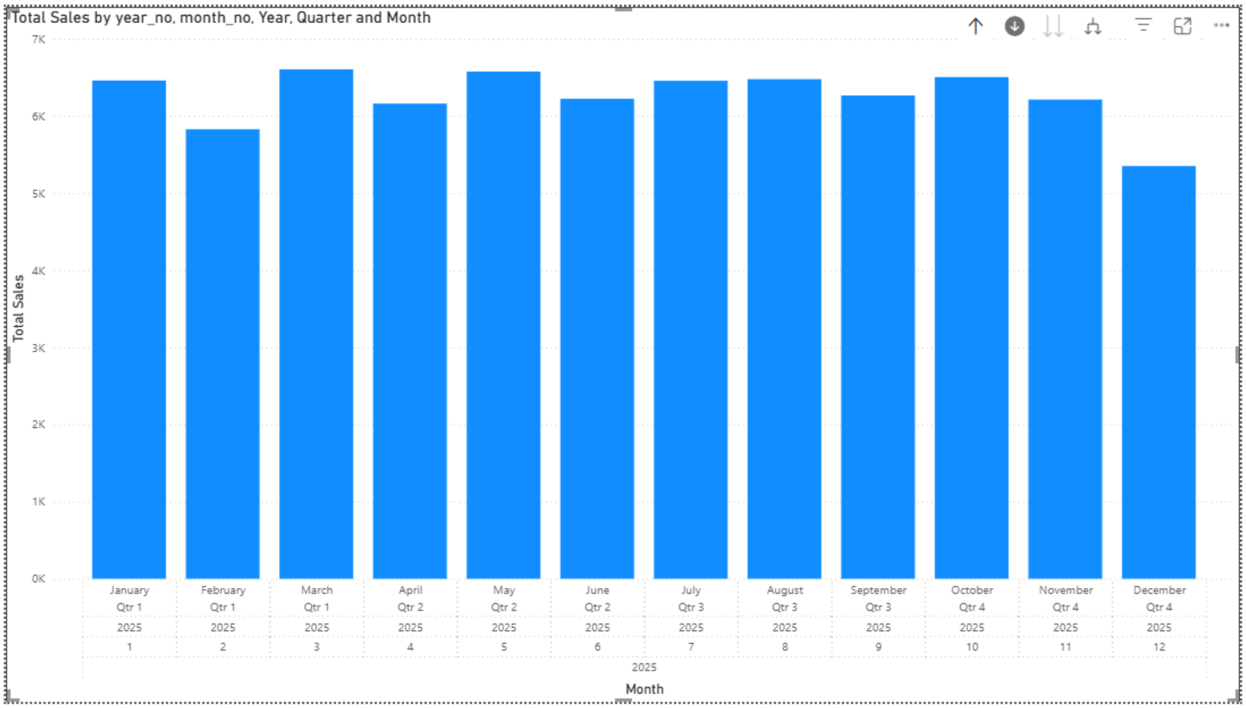
4.5. Report 3 — Drill-down report

A drill-down report was created using a clustered column chart with a hierarchy of Year, Month, and Full Date. Total Sales was used as the measure, allowing users to drill from summary level to detailed level.

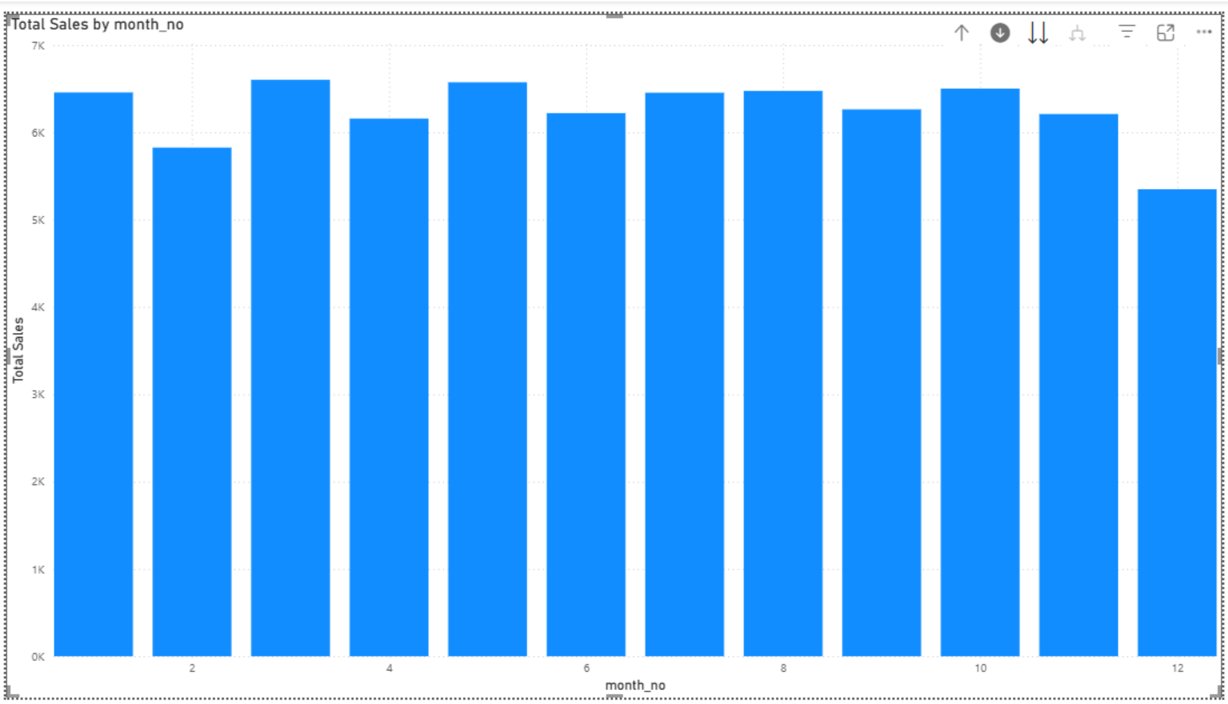
This report helps users move from yearly analysis into monthly and date-level detail to observe cinema sales trends more clearly.



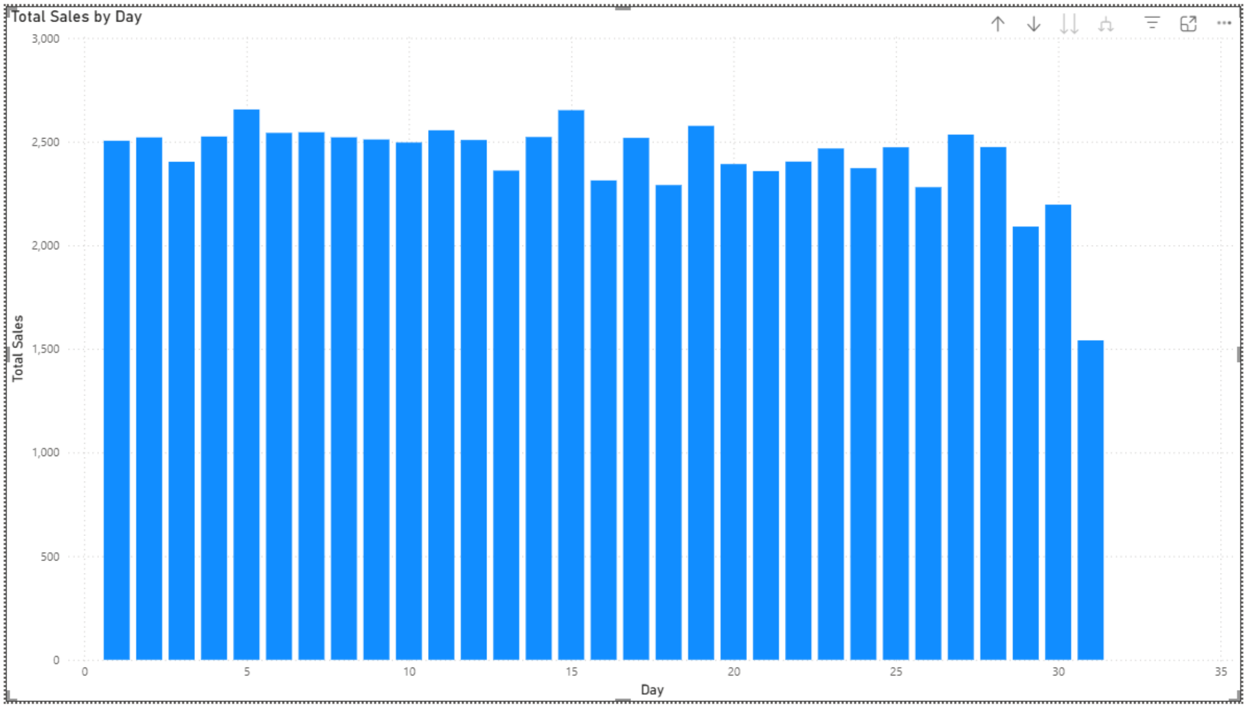
Drill-down chart at the Year level.



Total Sales by year_no, month_no, Year, Quarter and Month
7K
6K
5K
4K
3K
2K
1K
0K
Total Sales
January Qtr 1 2025 1
February Qtr 1 2025 2
March Qtr 1 2025 3
April Qtr 2 2025 4
May Qtr 2 2025 5
June Qtr 2 2025 6
July Qtr 3 2025 7
August Qtr 3 2025 8
September Qtr 3 2025 9
October Qtr 4 2025 10
November Qtr 4 2025 11
December Qtr 4 2025 12
2025
Month



Total Sales by month_no
7K
6K
5K
4K
3K
2K
1K
0K
Total Sales
2
4
6
8
10
12
month_no



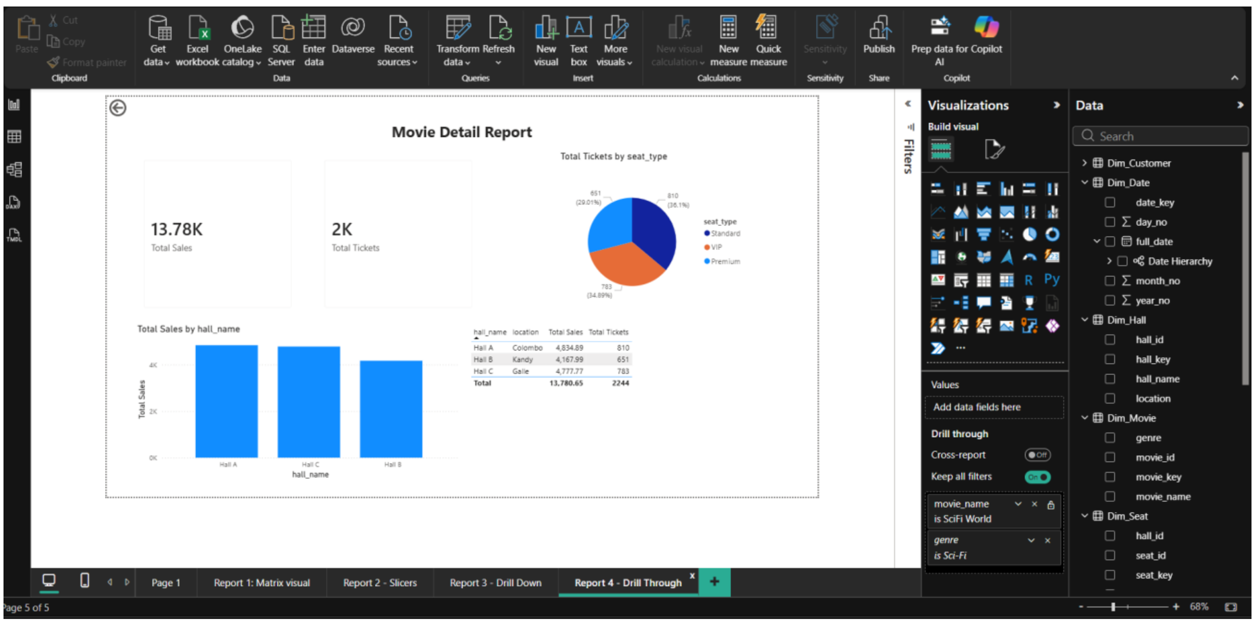
Drill-down chart expanded to Month or Full Date level.

4.6. Report 4- Drill-through report

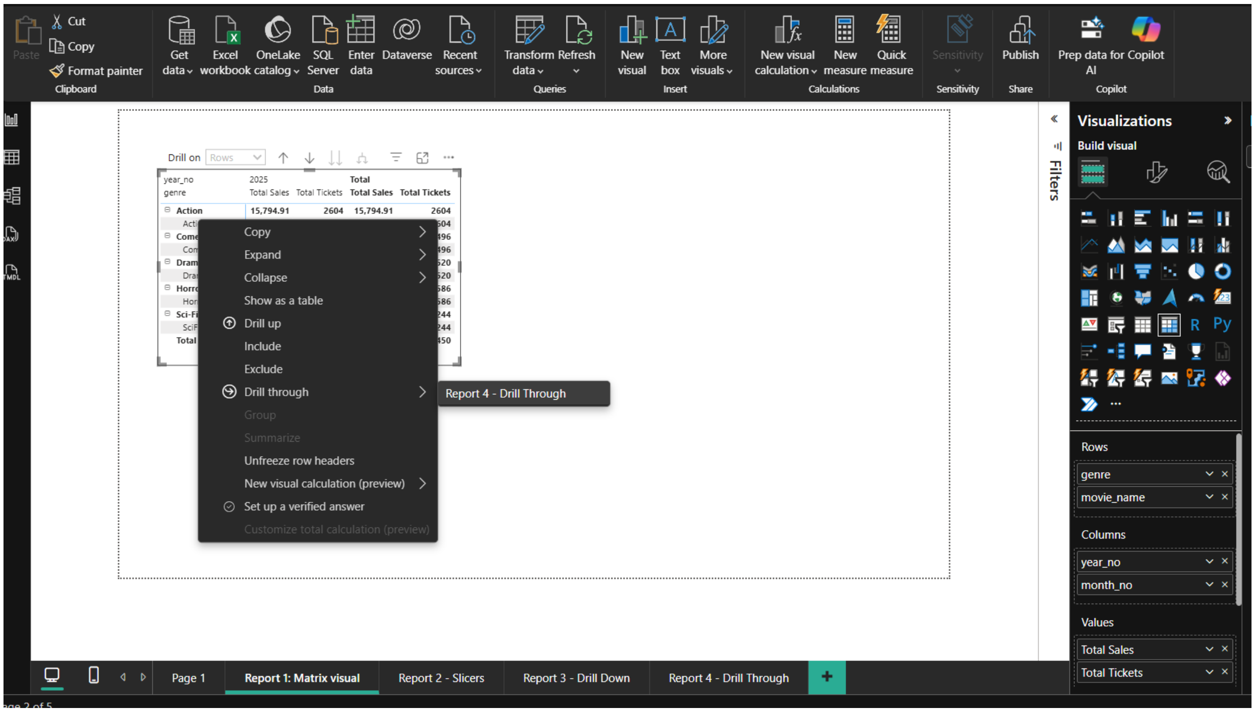
A drill-through page was created using movie_name as the drill-through field. The page includes cards, charts, and a table to show detailed information for the selected movie. A back button was added to return to the previous report page.

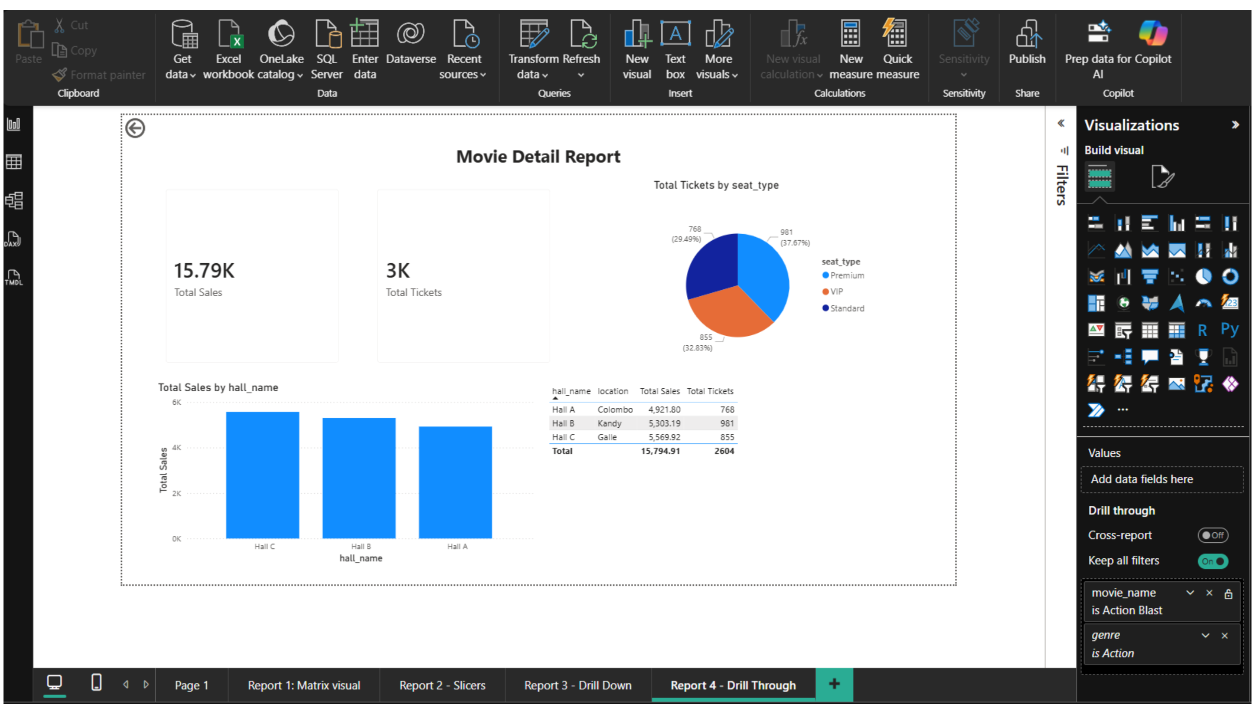
When a user right-clicks a movie in another report and navigates to this page, all visuals are filtered to show detailed information only for the selected movie.

These four reports together demonstrate tabular analysis, filtering, hierarchical exploration, and detailed navigation using Power BI.



Drill-through detail page design.

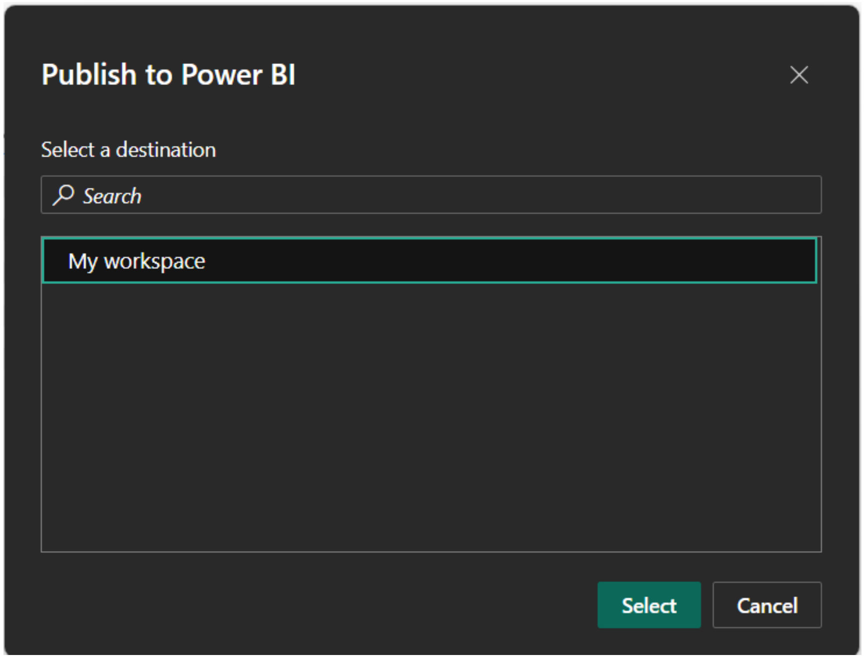


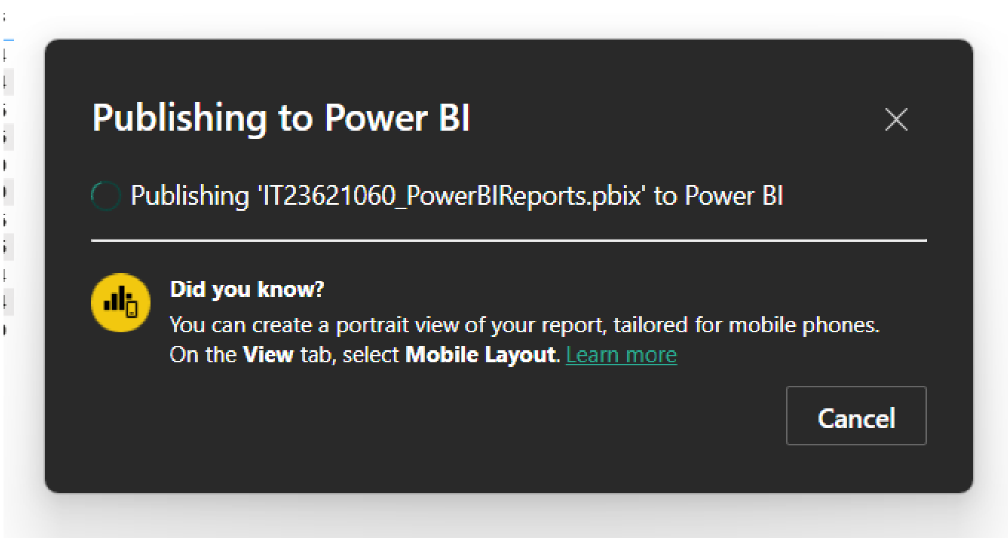


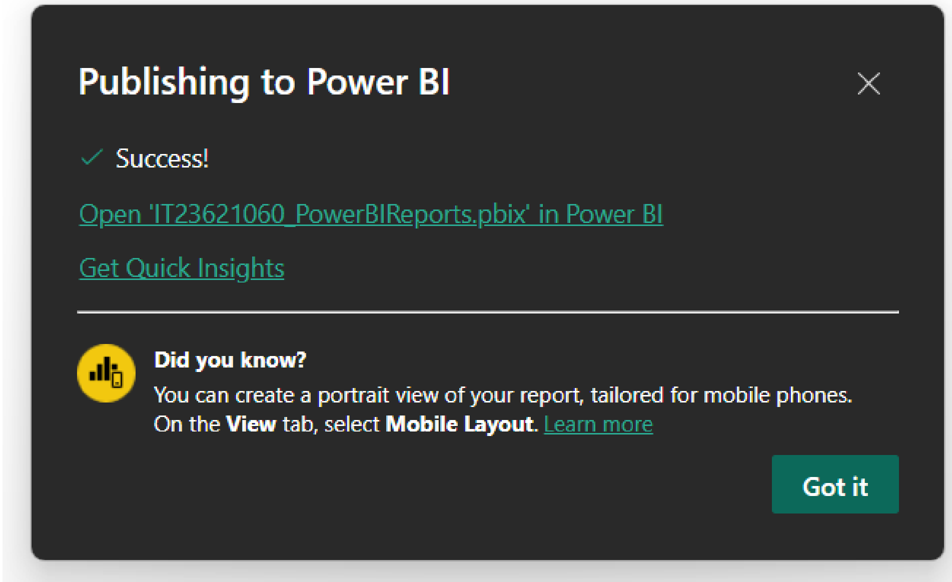
Filtered drill-through page for the selected movie.

Step 5 - Publishing to Power BI Service

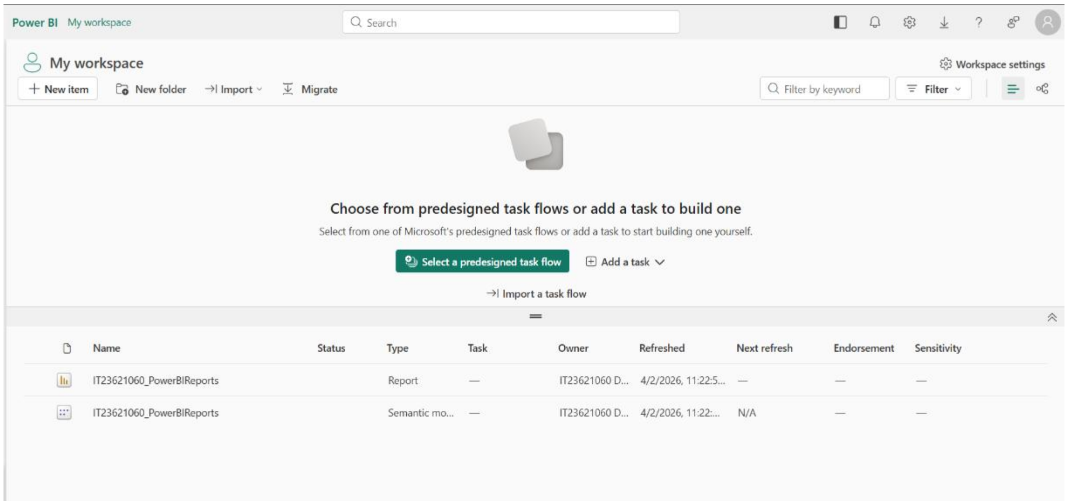
After completing the reports in Power BI Desktop, the .pbix file was published to Power BI Service using the Publish option. The report was uploaded to the online workspace and opened in the browser to confirm successful cloud deployment. This step satisfies the assignment requirement of demonstrating the reports in Power BI Service.

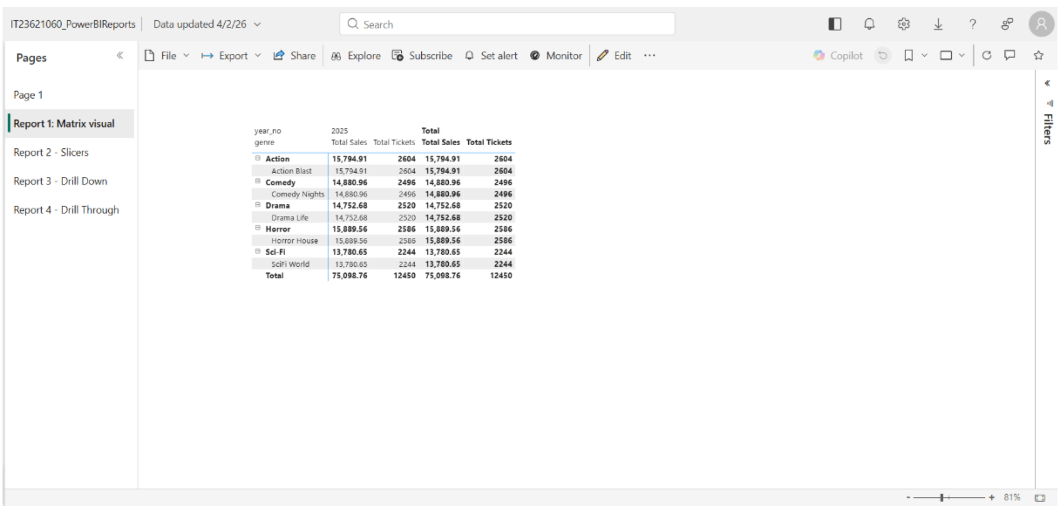


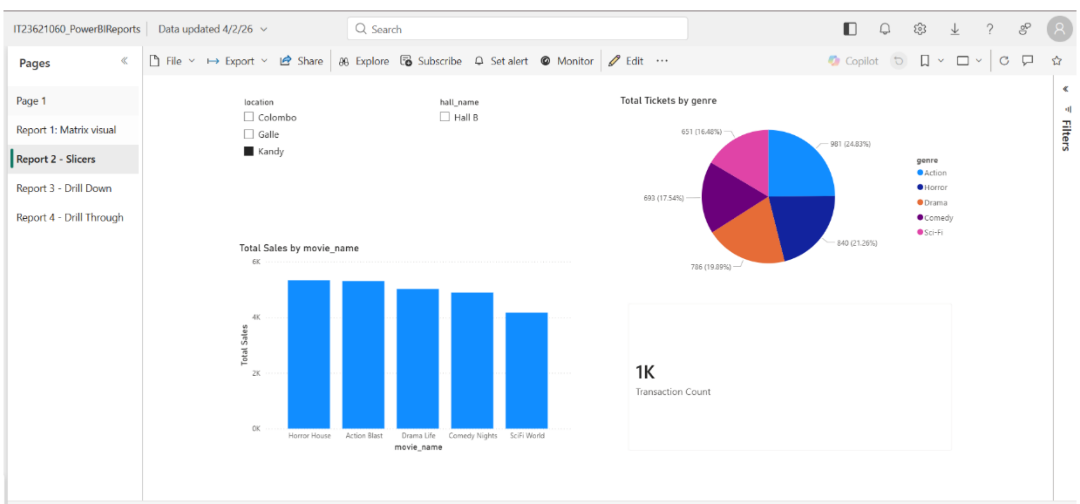


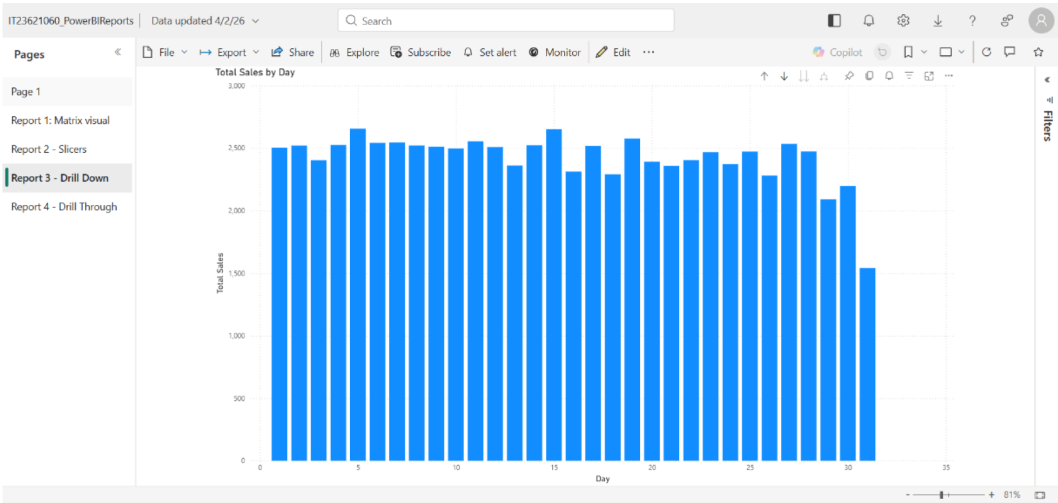


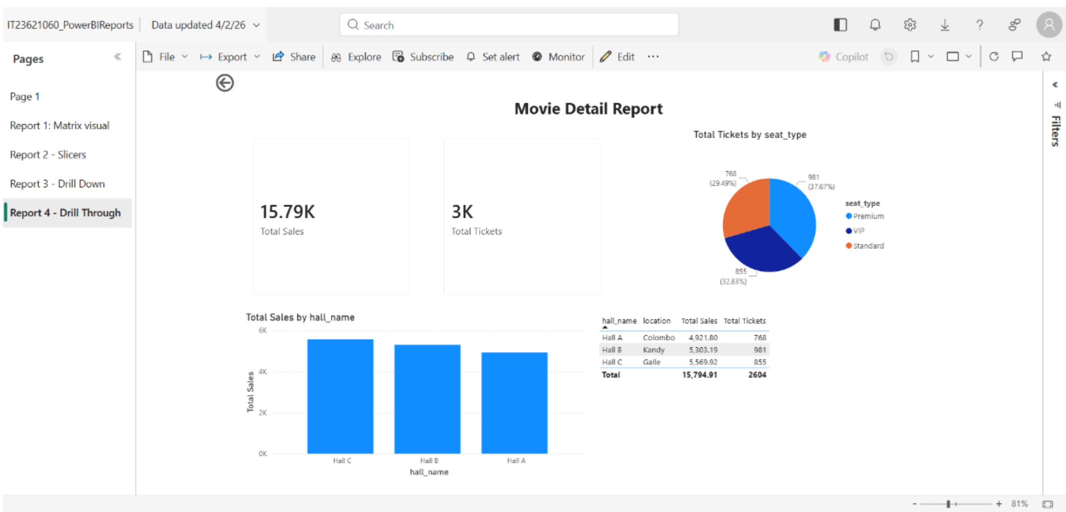
Successful publishing of the Power BI report to Power BI Service.


Power BI
My workspace
Search
My workspace
Workspace settings
New item
New folder
Import
Migrate
Filter by keyword
Filter
Choose from predesigned task flows or add a task to build one
Select from one of Microsoft's predesigned task flows or add a task to start building one yourself.
Select a predesigned task flow
Add a task
Import a task flow
Name
Status
Type
Task
Owner
Refreshed
Next refresh
Endorsement
Sensitivity
IT23621060_PowerBIReports
Report
—
IT23621060 D...
4/2/2026, 11:22:5...
—
—
—
IT23621060_PowerBIReports
Semantic mo...
—
IT23621060 D...
4/2/2026, 11:22:...
N/A
—
—


IT23621060_PowerBIReports
Data updated 4/2/26
Search
Pages
File
Export
Share
Explore
Subscribe
Set alert
Monitor
Edit
Copilot
Page 1
Report 1: Matrix visual
Report 2 - Slicers
Report 3 - Drill Down
Report 4 - Drill Through
Filters
year_no
genre
2025
Total
Total Sales
Total Tickets
Total Sales
Total Tickets
Action
15,794.91
2604
15,794.91
2604
Action Blast
15,794.91
2604
15,794.91
2604
Comedy
14,880.96
2496
14,880.96
2496
Comedy Nights
14,880.96
2496
14,880.96
2496
Drama
14,752.68
2520
14,752.68
2520
Drama Life
14,752.68
2520
14,752.68
2520
Horror
15,889.56
2586
15,889.56
2586
Horror House
15,889.56
2586
15,889.56
2586
Sci-Fi
13,780.65
2244
13,780.65
2244
SciFi World
13,780.65
2244
13,780.65
2244
Total
75,098.76
12450
75,098.76
12450
81%


IT23621060_PowerBIReports
Data updated 4/2/26
Search
Pages
File
Export
Share
Explore
Subscribe
Set alert
Monitor
Edit
Copilot
Page 1
Report 1: Matrix visual
Report 2 - Slicers
Report 3 - Drill Down
Report 4 - Drill Through
Filters
location
Colombo
Galle
Kandy
hall_name
Hall B
Total Tickets by genre
651 (16.46%)
981 (24.83%)
693 (17.54%)
840 (21.26%)
786 (19.89%)
genre
Action
Horror
Drama
Comedy
Sci-Fi
Total Sales by movie_name
6K
4K
2K
0K
Total Sales
Horror House
Action Blast
Drama Life
Comedy Nights
SciFi World
movie_name
1K
Transaction Count
81%





Published report viewed in Power BI Service (online).

Conclusion

The Assignment 1 data warehouse was successfully extended into a complete BI solution for Assignment 2. An SSAS cube was implemented with measures, dimensions, and a date hierarchy. OLAP operations were demonstrated in Excel using PivotTables, and interactive reports were created in Power BI Desktop and published to Power BI Service. The final solution supports meaningful analysis of cinema ticket sales and customer behavior.

References:

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- Microsoft. *Power BI documentation*. Available at: https://learn.microsoft.com/en-us/power-bi/
- Kimball Group. *Dimensional Modeling Techniques*. Available at: https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/kimball-techniques/dimensional-modeling-techniques/